ASSIGNMENT 2 FRONT SHEET

Qualification	**BTEC Level 5 HND Diploma in Business**		
Unit number and title	**Unit 30: Application Development**		
Submission date	23/10/2022	**Date Received 1st submission**	
Re-submission Date		**Date Received 2nd submission**	
Student Name	Le Hoang Du	**Student ID**	GCC200416
Class	GCC0901	**Assessor name**	Pham Truong Hong Ngan
Student declaration I certify that the assignment submission is entirely my own work and I fully understand the consequences of plagiarism. I understand that making a false declaration is a form of malpractice.			
		Student's signature	Le Hoang Du

Grading grid

P4	P5	P6	M3	M4	M5	D2	D3

❒ **Summative Feedback:**		❒ **Resubmission Feedback:**
Grade:	**Assessor Signature:**	**Date:**
Internal Verifier's Comments:		
Signature & Date:		

Assignment Brief 2 (RQF)

Higher National Certificate/Diploma in Computing

Student Name/ID Number:	**Le Hoang Du/GCC200416**
Unit Number and Title:	**Unit 30: Application Development**
Academic Year:	**2021 – 2022**
Unit Assessor:	**Pham Truong Hong Ngan**
Assignment Title:	**Application development with design diagrams and code**
Issue Date:	
Submission Date:	**23/10/2022**
Internal Verifier Name:	
Date:	

Submission Format:

Format:

- An individual report document in PDF

Submission

- Students are compulsory to submit the assignment in due date and in a way requested by the Tutor.
- The form of submission will be a soft copy posted on http://cms.greenwich.edu.vn/.
- Remember to convert the word file into PDF file before the submission on CMS.

Note:

- The individual Assignment *must* be your own work, and not copied by or from another student.
- If you use ideas, quotes or data (such as diagrams) from books, journals or other sources, you must reference your sources, using the Harvard style.
- Make sure that you understand and follow the guidelines to avoid plagiarism. Failure to comply this requirement will result in a failed assignment.

Unit Learning Outcomes:

LO3 Work individually and as part of a team to plan and produce a functional business application with support documentation

LO4 Evaluate the performance of a business application against its Software Design Document and initial requirements

Assignment Brief and Guidance:

Assignment scenario (continued from Assignment 1) Your team has finished the analysis and design for the system. Next task is development of the system.

Tasks:

After the presentation about your design (from Assignment 1), you need to create a formal questionnaire that effectively reviews your business application, problem definition statement, proposed solution and development strategy. This formal questionnaire should be answered by your colleagues. For any new insights, ideas or potential improvements to your system you need to evaluate and justify the reasons why you have chosen to include (or not to include) them as part of this business application. Based on the feedback of your colleagues, amend the design if needed.

Next task is to develop the business application based on the design, chosen technologies and methodology. When the application is fully built and tested, you need to review its performance against the Software Requirement Specificationn, analyze the factors that influence its performance and use them to undertake a critical review of the design, development and testing stages of your application. Conclude your review by reflectively discussing your previously identified risks. You should evaluate the strengths and weaknesses of your business application and fully justify opportunities for improvement and further development.

To conclude, your report document should include:

- Peer review section (questionnaire and answers, your reflection on the feedback)
- Development section (how you develop and test the application, what is the result)
- Review section (review, analyse and critical evaluate your application)

Your team needs to prepare a demo based on this report for the final demonstration.
The working application must also be demonstrated.

Learning Outcomes and Assessment Criteria (Assignment 2):			
Learning Outcome	Pass	Merit	Distinction
LO3	**P4** Create a formal questionnaire that effectively reviews your business application, problem definition statement, proposed solution and development strategy. Use this questionnaire as part of a peer-review and document any feedback given. **P5** Develop a functional business application based on a specified business problem.	**M3** Interpret your peer-review feedback and identify opportunities not previously considered. **M4** Develop a functional business application based on a specific Software Design Document with supportive evidence of using the preferred tools, techniques and methodologies.	**D2** Evaluate any new insights, ideas or potential improvements to your system and justify the reasons why you have chosen to include (or not to include) them as part of this business application.
LO4	**P6** Review the performance of your business application against the Problem Definition Statement and initial requirements.	**M5** Analyse the factors that influence the performance of a business application and use them to undertake a critical review of the design, development and testing stages of your application. Conclude your review by reflectively discussing your previously identified risks.	**D3** Critically evaluate the strengths and weaknesses of your business application and fully justify opportunities for improvement and further development.

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2.4 Screenshots of using GitHub or GitLab to manage the source code

GitHub is a platform for developers to exchange knowledge and source code. That's why we decided to use GitHub in this project to easily exchange the source code with each other if there is any change in the project. Here is a picture of our repository on the GitHub platform:

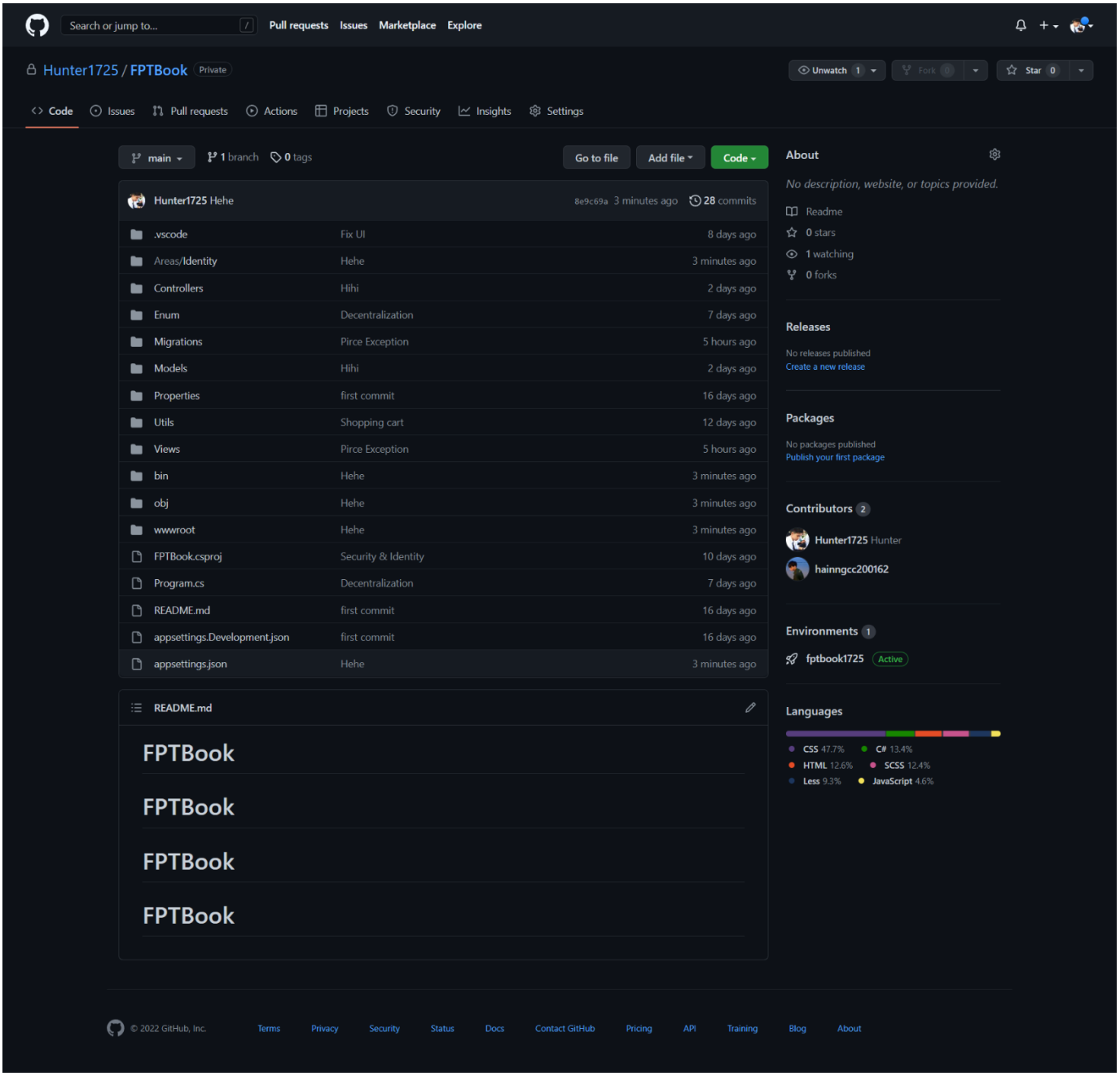


Figure 65: Repository on GitHub

Here are a few proof images of the source code repository in our project throughout the process that we code and manage the source code on the GitHub platform. For example, we can see below each member including Le Hoang Du, and Nguyen Ngoc Hai commit on October 18, October 17, October 16, October 15, etc.

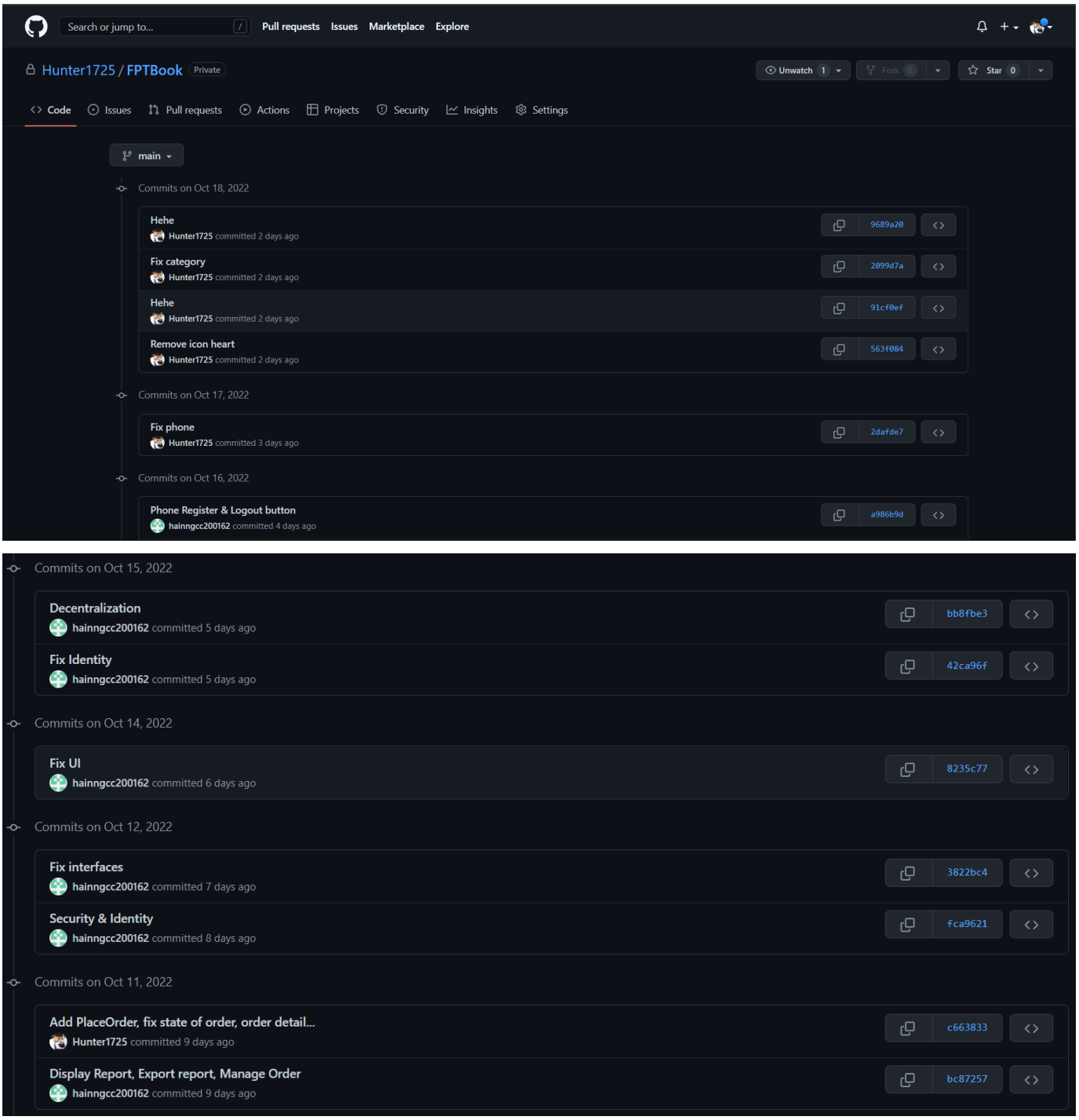


Figure 66: Some images about our commits on the GitHub

2.5 Screenshots of using IIS or Azure for the application deployment

Internet Information Services (IIS) for Windows® Server is a flexible, secure and manageable Web server for hosting anything on the Web. From media streaming to web applications, IIS's scalable and open architecture is ready to handle the most demanding tasks. Therefore, we have decided to choose IIS to deploy to our website. Here are the steps we took to deploy the site on IIS:

Step 1: Turn on Internet Information Service feature of the window OS:

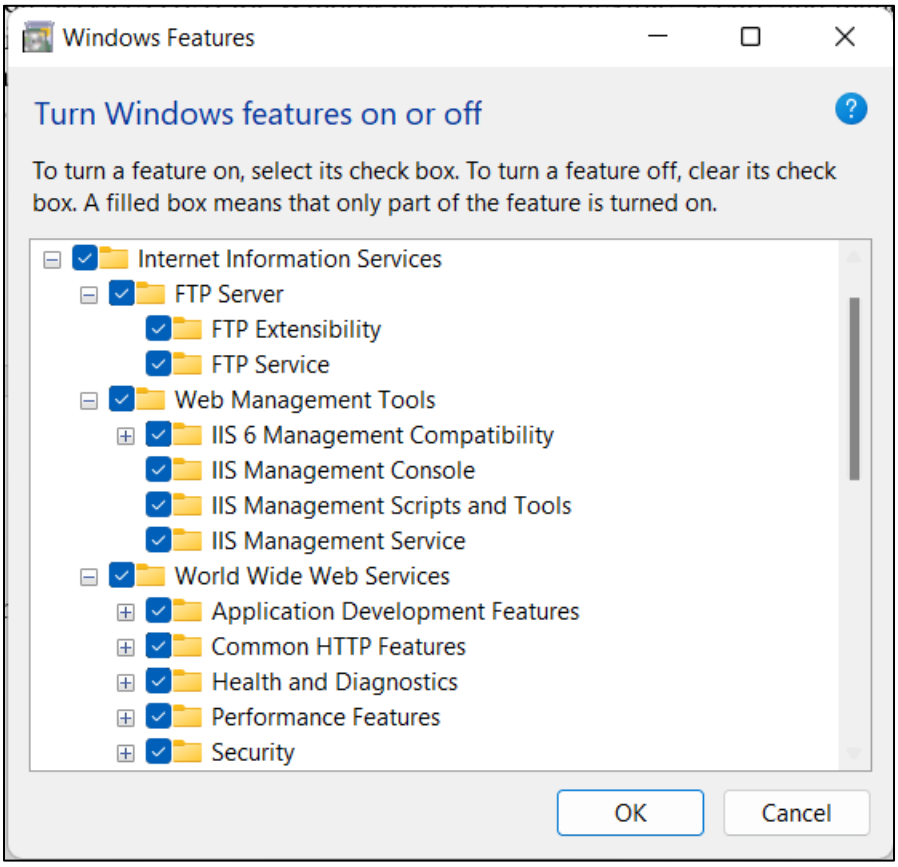


Figure 67: Enable IIS

Step 2: Download and install the .NET Core Hosting Bundle on Windows Server:

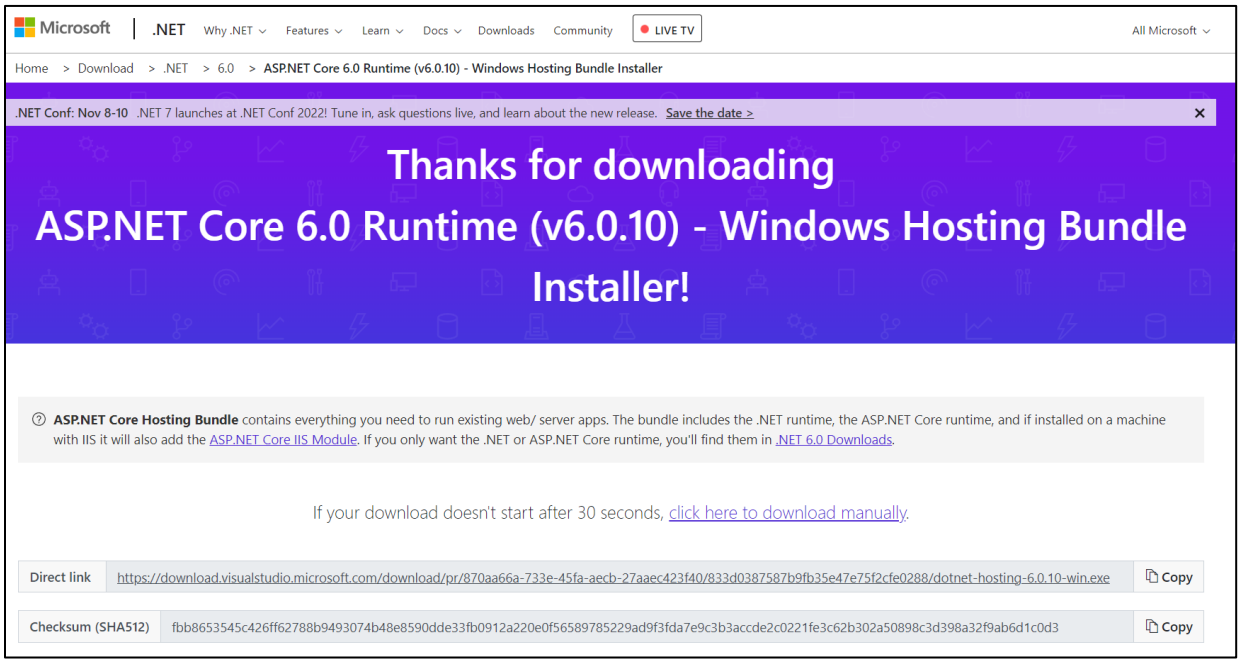


Figure 68: Download and install the .NET Core Hosting Bundle

Step 3: Restart the server or execute net stop was /y followed by net start w3svc in a command shell

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> net stop was /y
The following services are dependent on the Windows Process Activation Service service.
Stopping the Windows Process Activation Service service will also stop these services.

   World Wide Web Publishing Service

The World Wide Web Publishing Service service is stopping.
The World Wide Web Publishing Service service was stopped successfully.

The Windows Process Activation Service service is stopping.
The Windows Process Activation Service service was stopped successfully.

PS C:\WINDOWS\system32> net start w3svc
The World Wide Web Publishing Service service is starting.
The World Wide Web Publishing Service service was started successfully.

PS C:\WINDOWS\system32>
```

Figure 69: Execute on command shell

Step 4: Publish the app in Release configuration with the dotnet publish command:

```
PS C:\CodeApplicationDevelopment\FPTBook> dotnet publish --configuration Release
MSBuild version 17.3.2+561848881 for .NET
  Determining projects to restore...
  All projects are up-to-date for restore.
```

```
  FPTBook -> C:\CodeApplicationDevelopment\FPTBook\bin\Release\net6.0\FPTBook.dll
  FPTBook -> C:\CodeApplicationDevelopment\FPTBook\bin\Release\net6.0\publish\
PS C:\CodeApplicationDevelopment\FPTBook>
```

Figure 70: Publish the app

Step 5: Move the contents of the bin/Release/net6.0/publish folder to the IIS site folder on the server, which is the site's Physical path in IIS Manager.

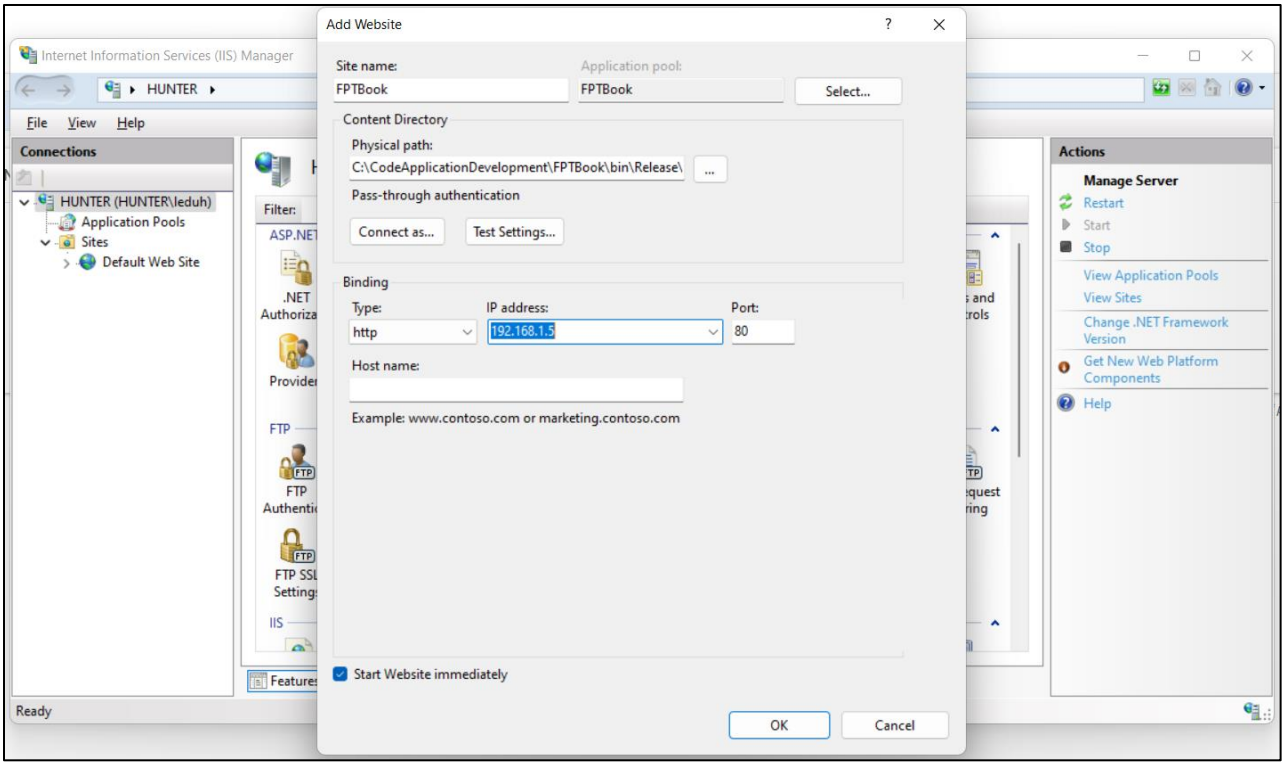


Figure 71: Add new site

Step 6: Browse the website with the IP selected when creating a new site at IIS manager

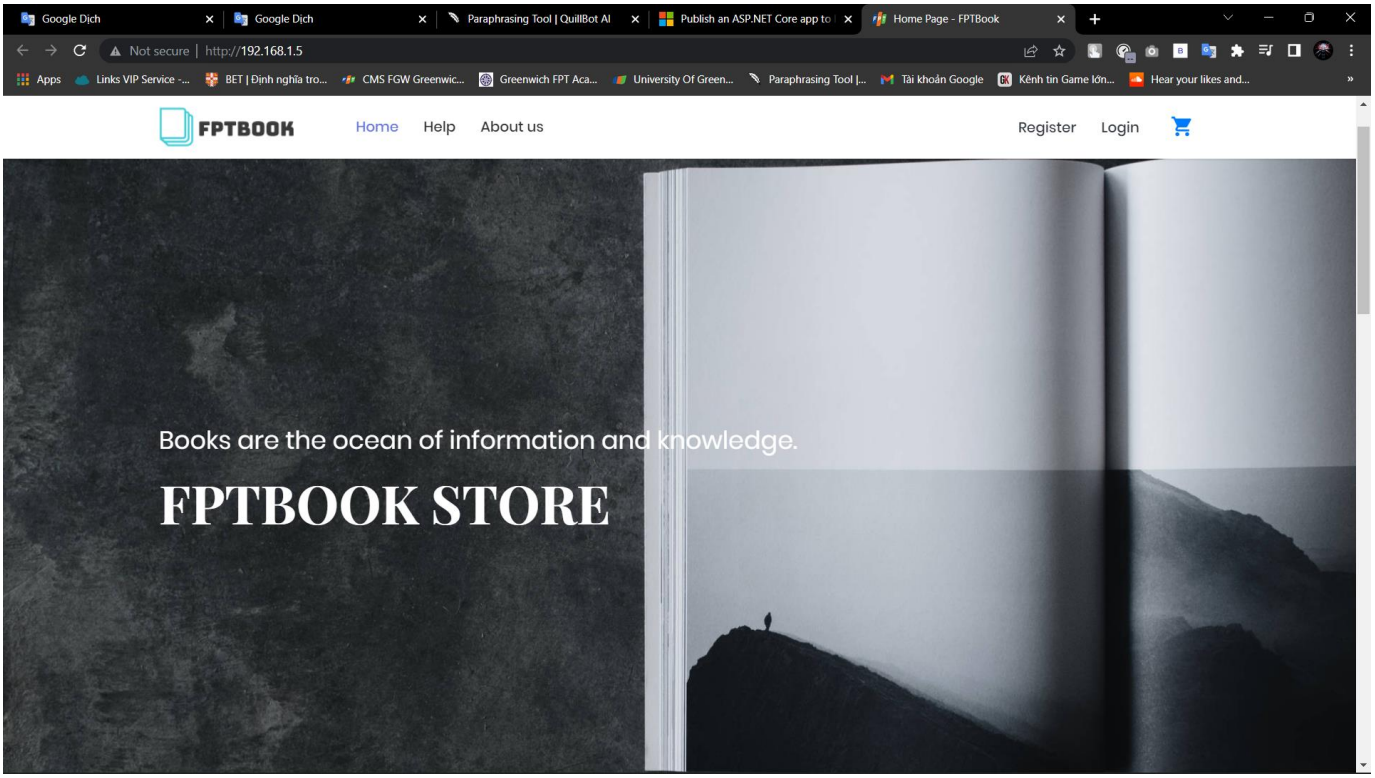
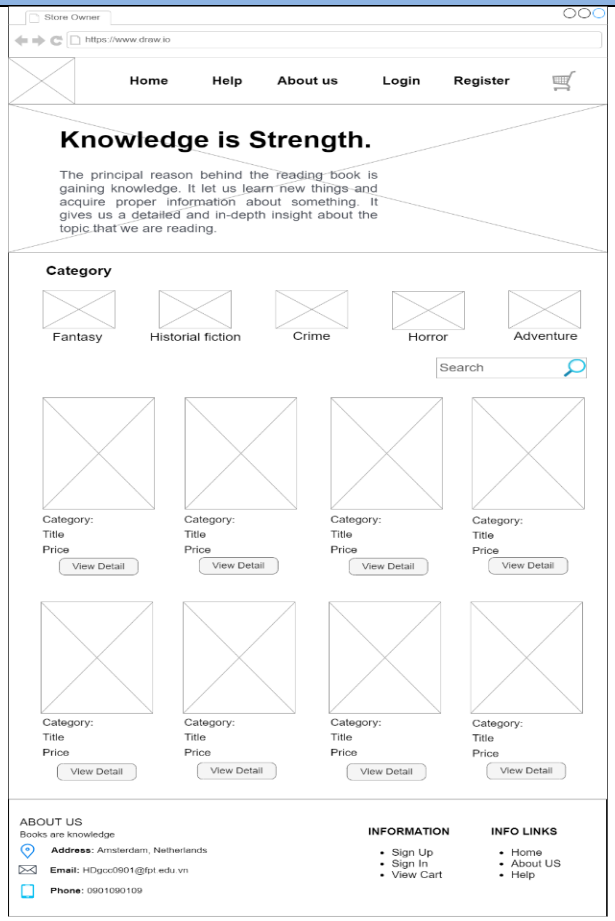
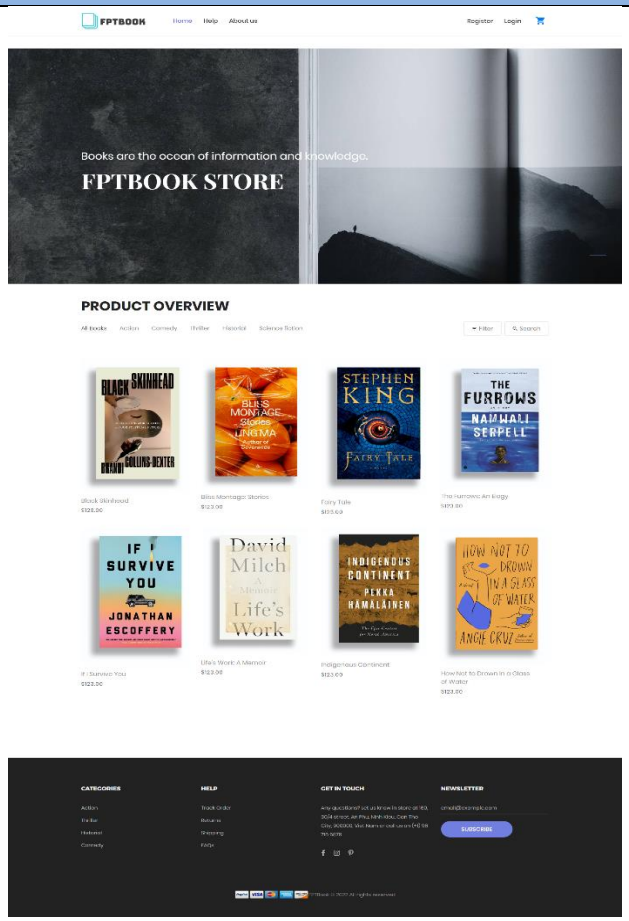


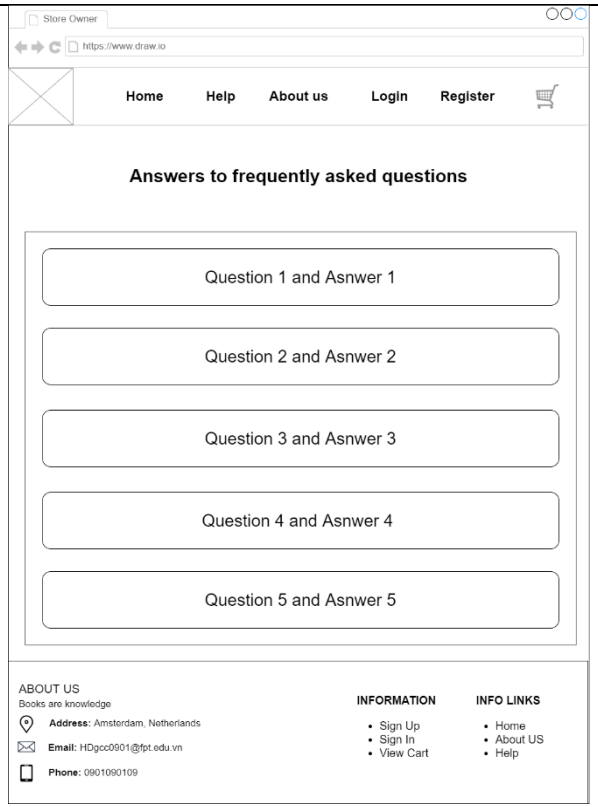
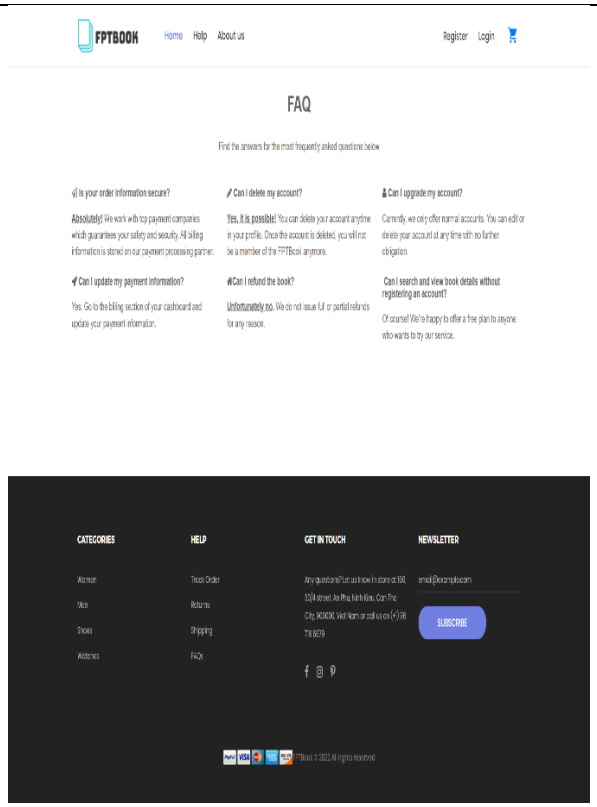
Figure 72: Browse the website

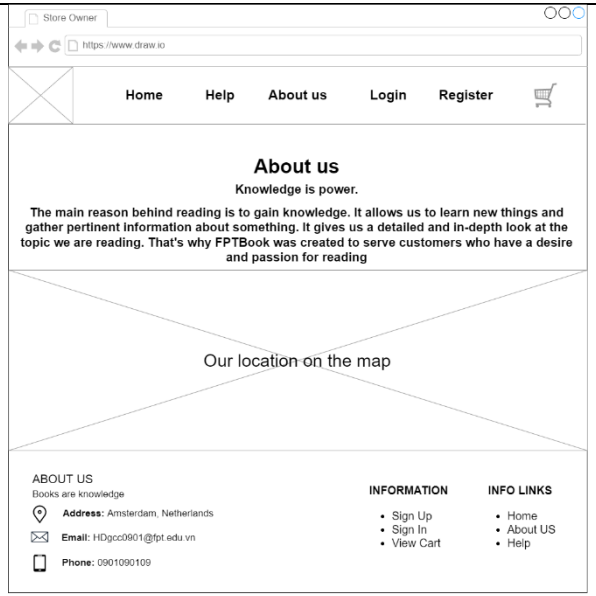
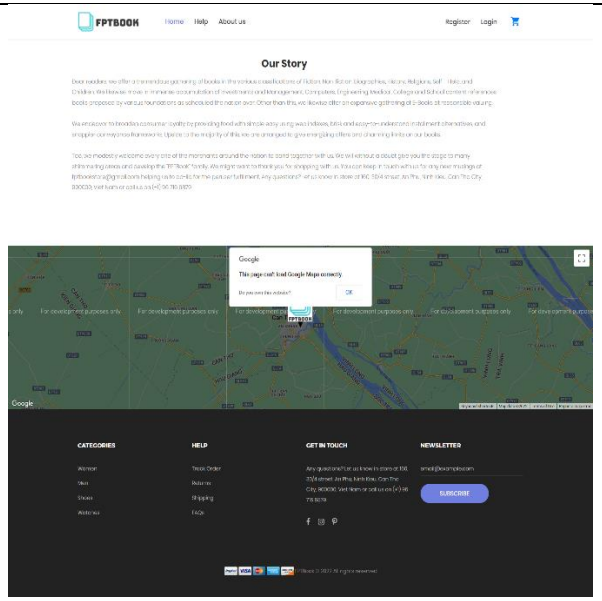
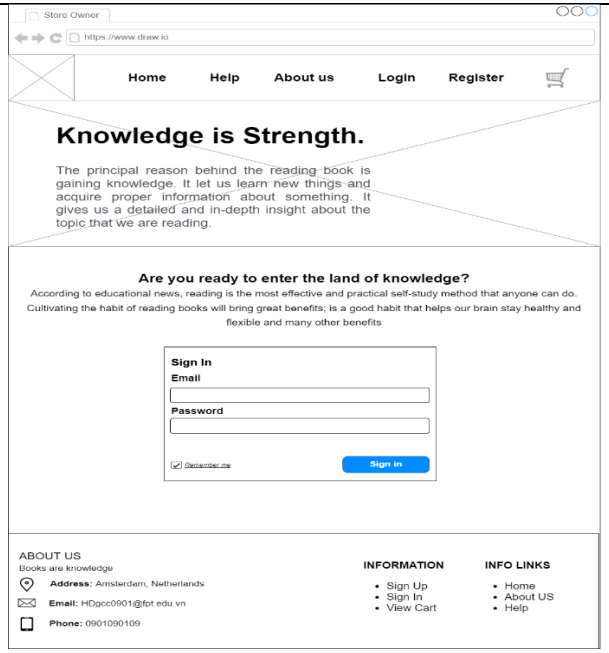
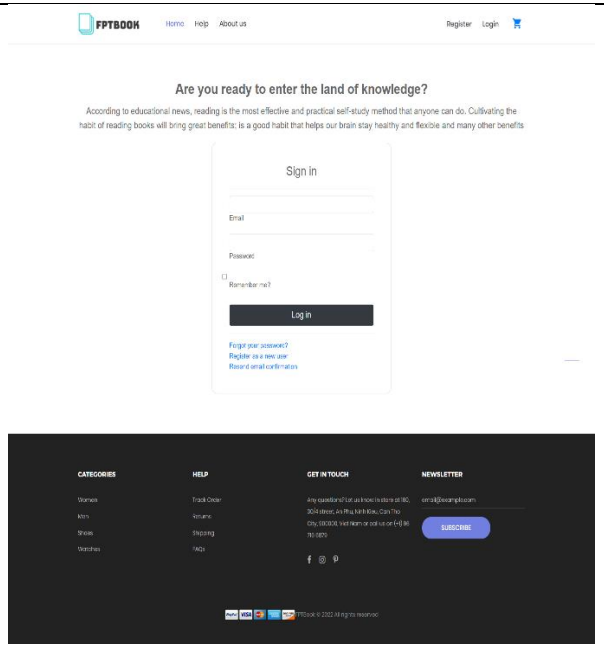
3 Application Evaluation

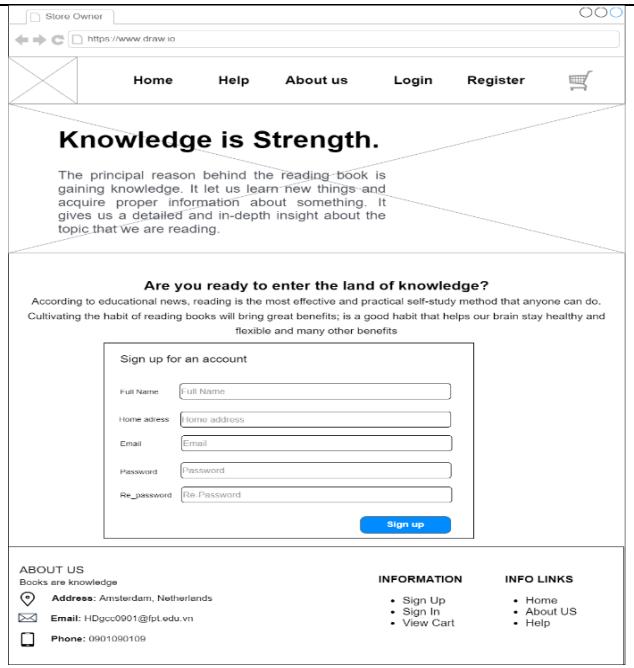
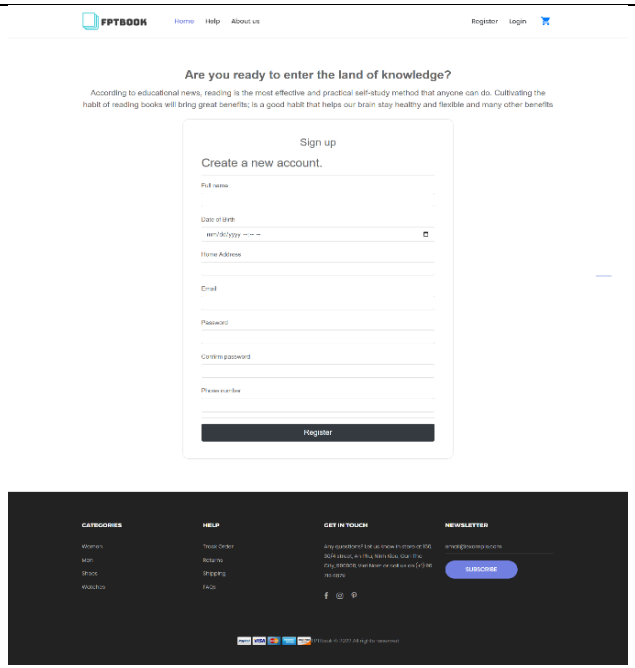
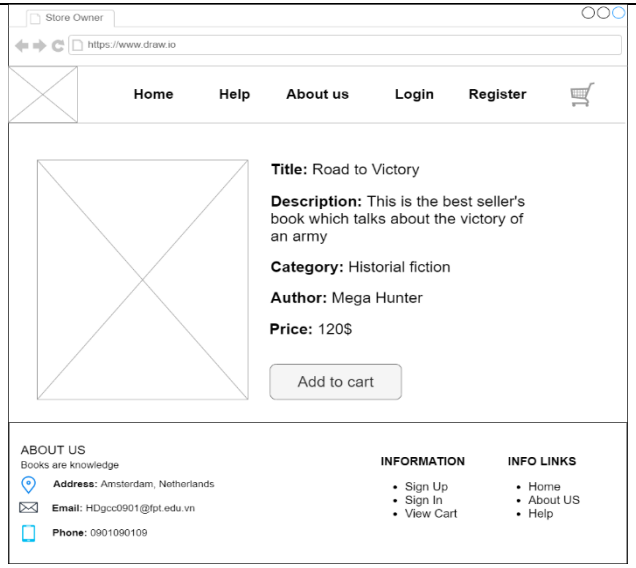
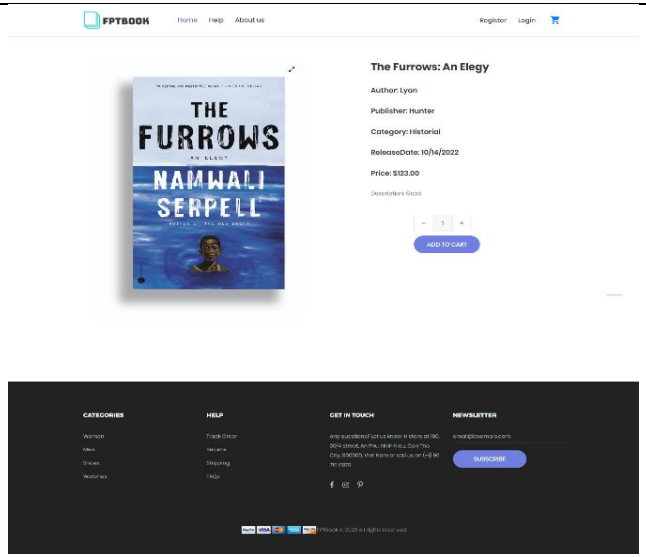
3.1 Review the performance of the application

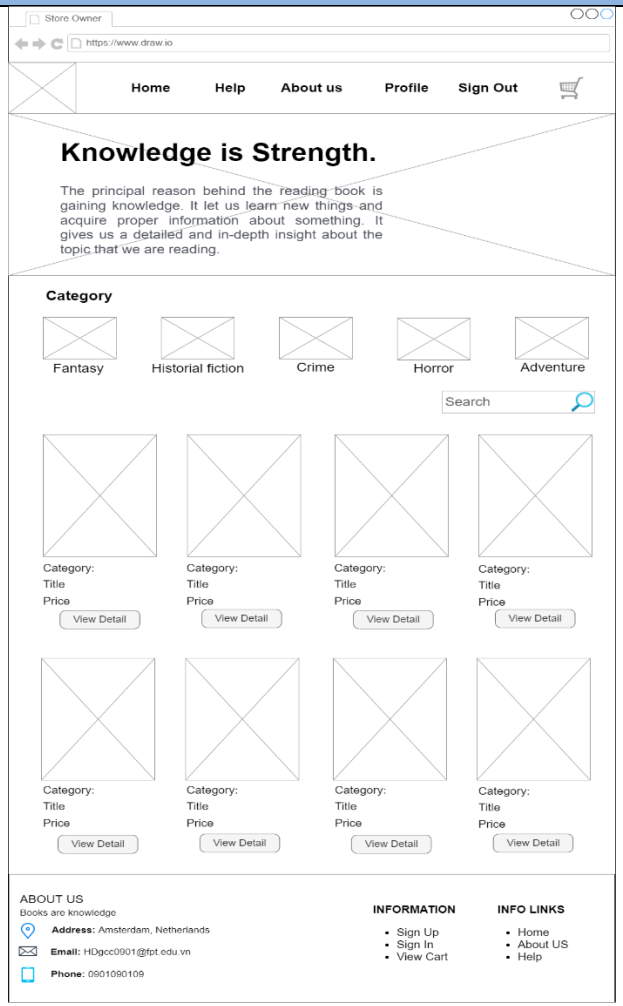
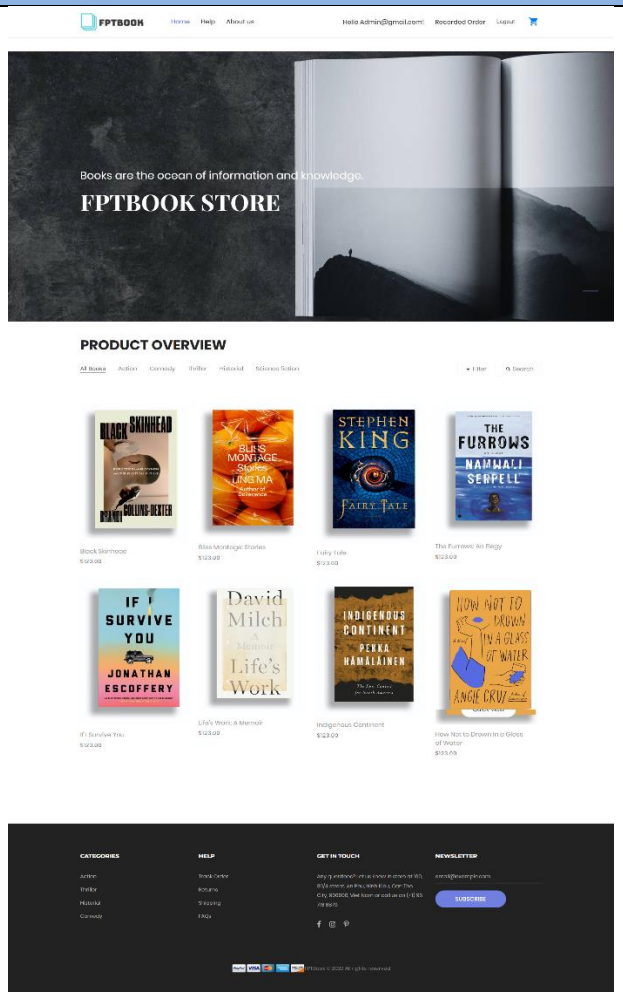
Table 2: Compare wireframe vs finale result

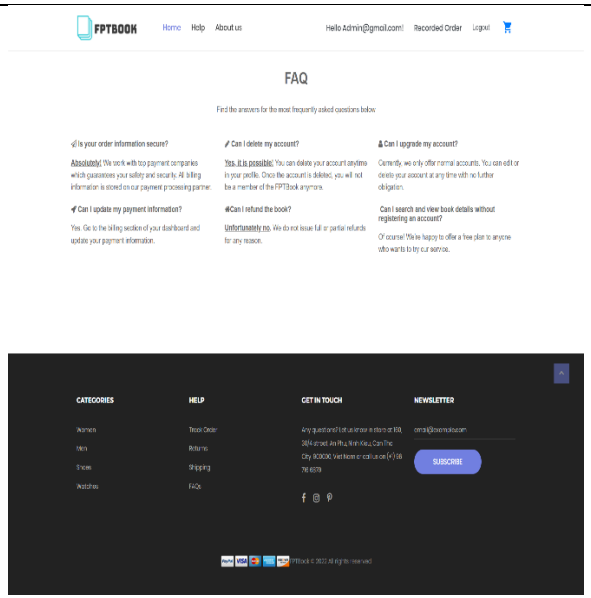
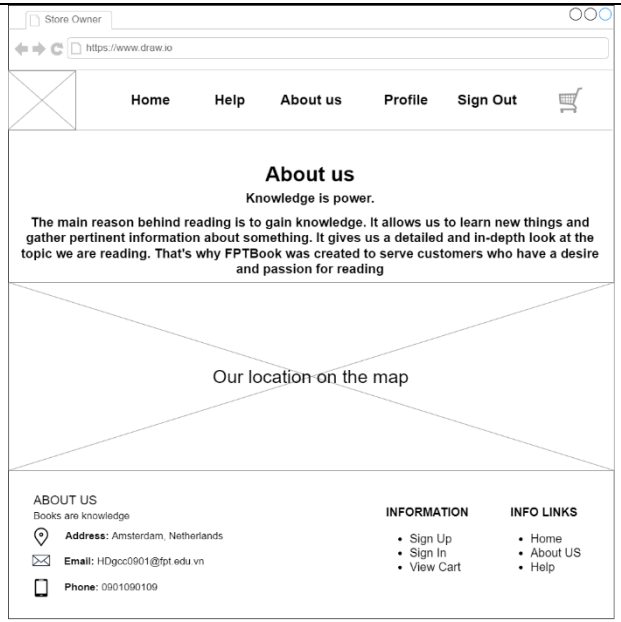
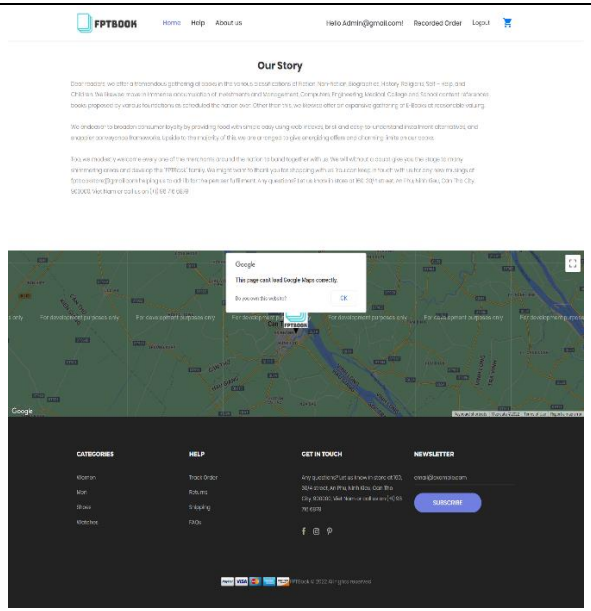
No.	Describe	Wireframe	Result	Evaluation
Unauthenticated User Role				
1	Home page			The interface of the website has had some minor changes compared to the original wireframe design, these changes make the user interface more beautiful and smoother than before.

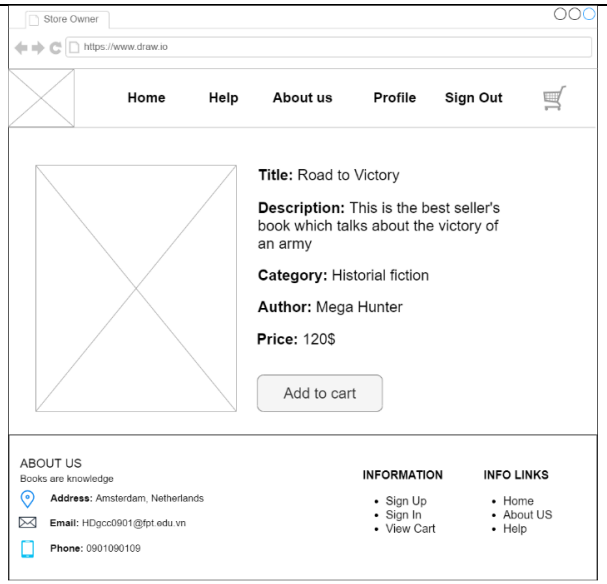
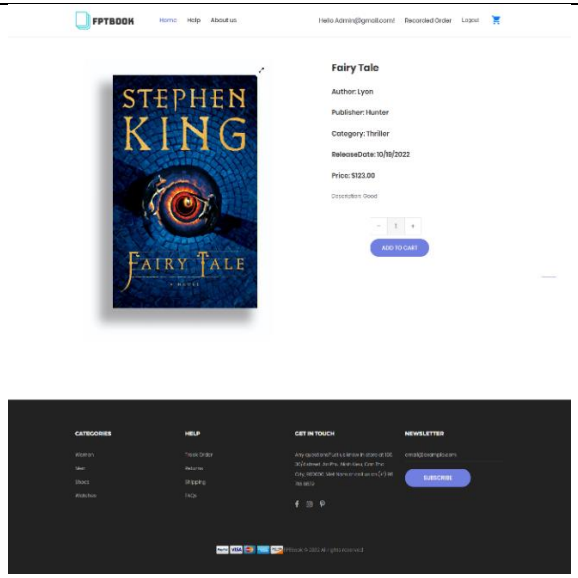
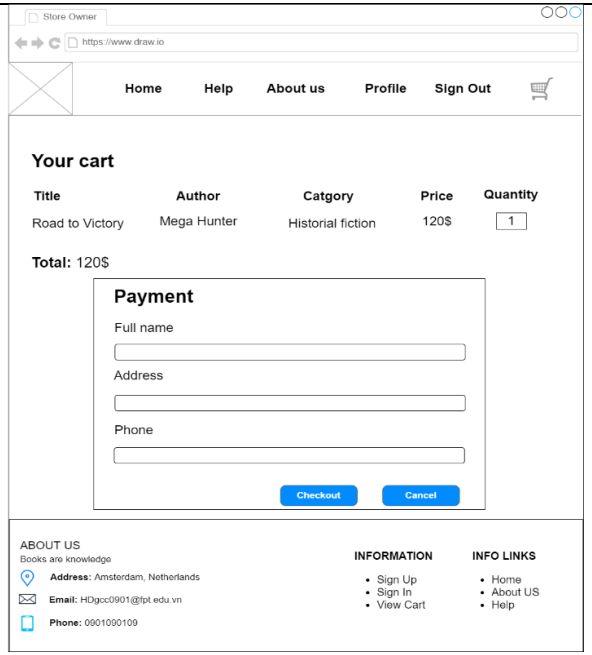
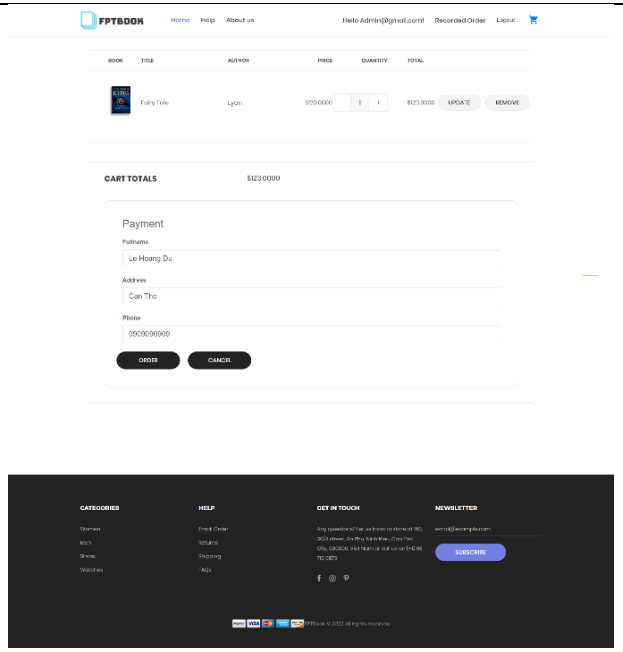
2	Help page			The interface of the website has had some minor changes compared to the original wireframe design, these changes make the user interface more beautiful and smoother than before.

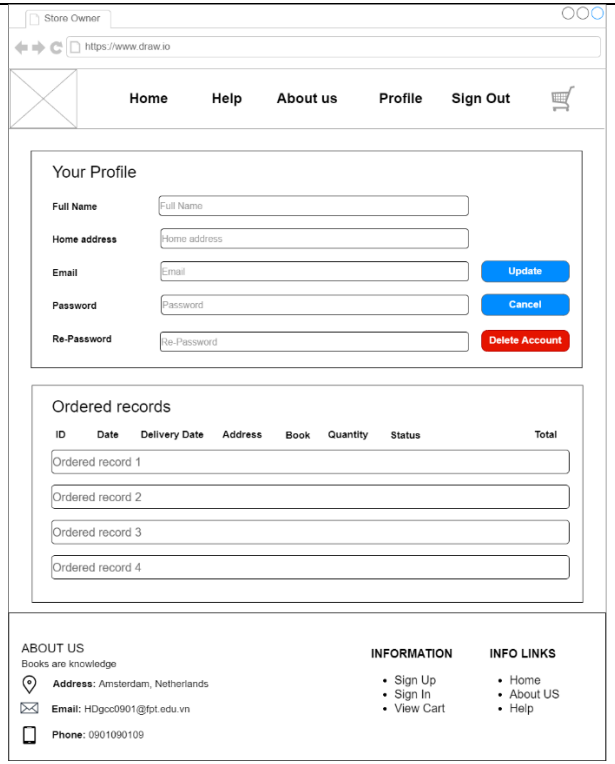
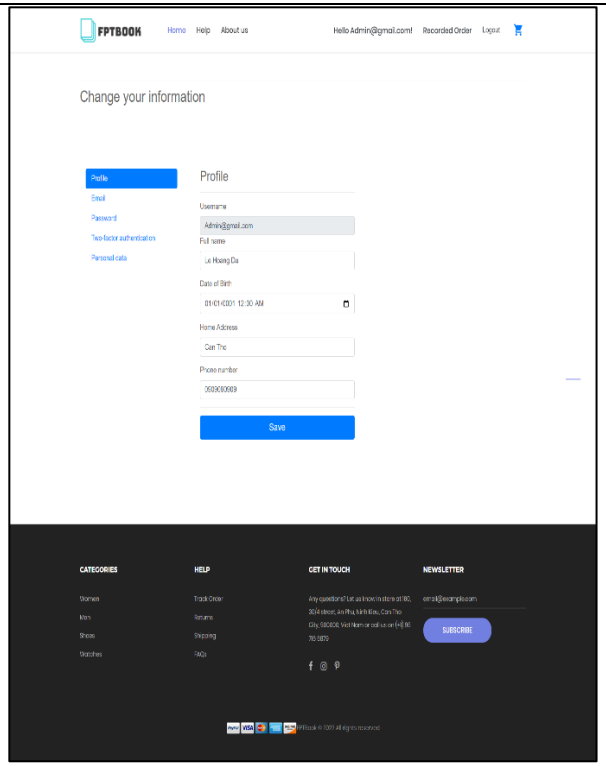
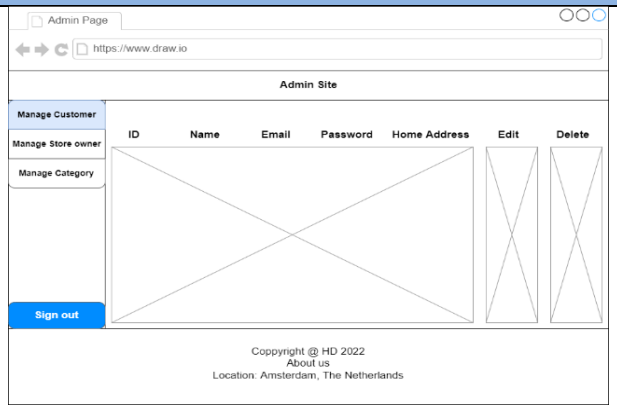
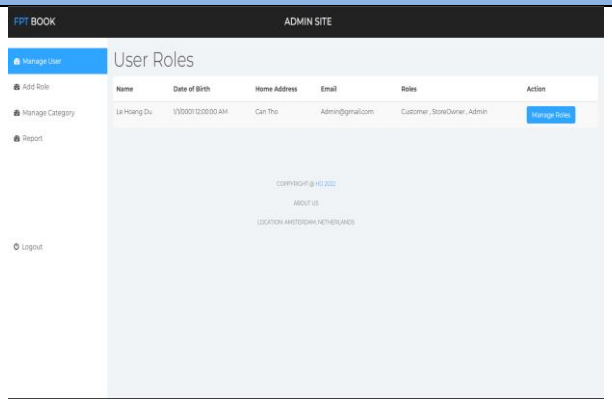
3	About us page			Wireframe and website screenshots are mostly the same. The website interface is designed based on pre-design wireframes.
4	Sign in page			The interface of the website has had some minor changes compared to the original wireframe design, these changes make the user interface more beautiful and smoother than before.

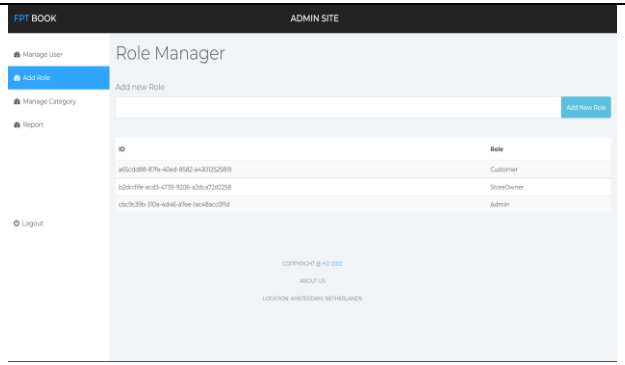
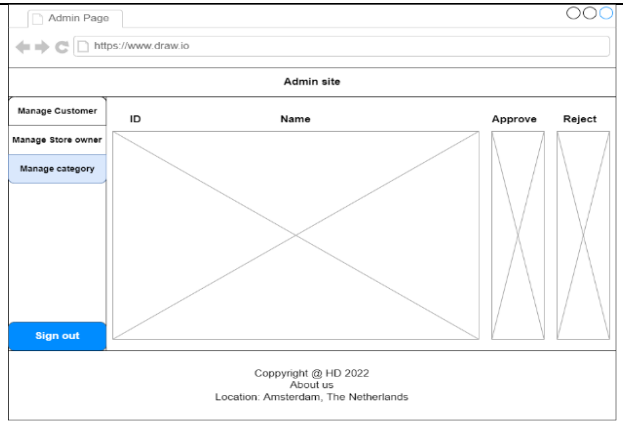
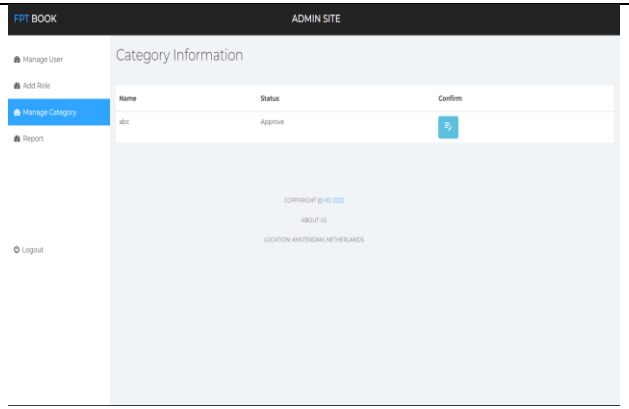
5	Register page			The interface of the website has had some minor changes compared to the original wireframe design, these changes make the user interface more beautiful and smoother than before.
6	View Detail page			Wireframe and website screenshots are mostly the same. The website interface is designed based on pre-design wireframes.

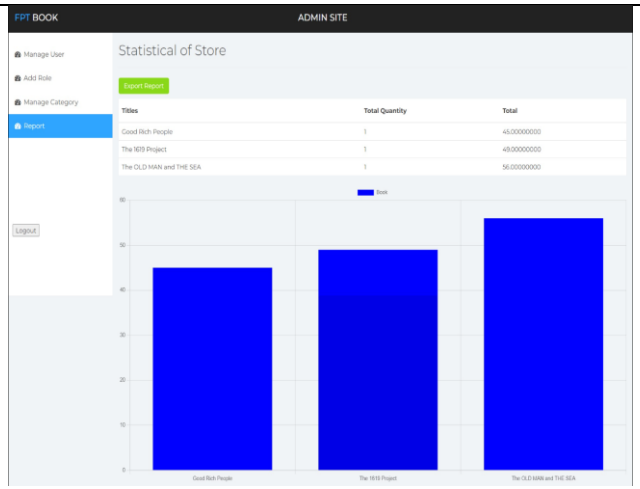
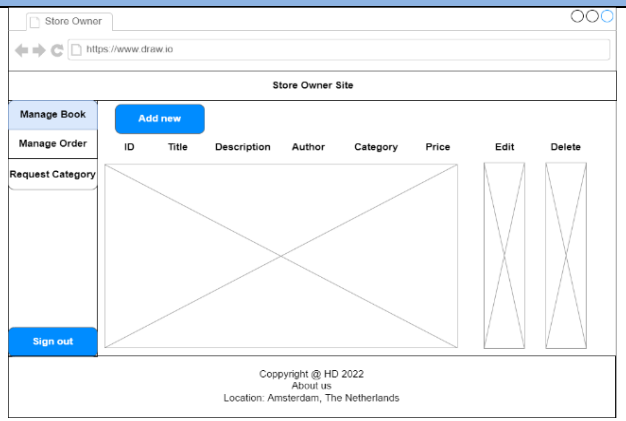
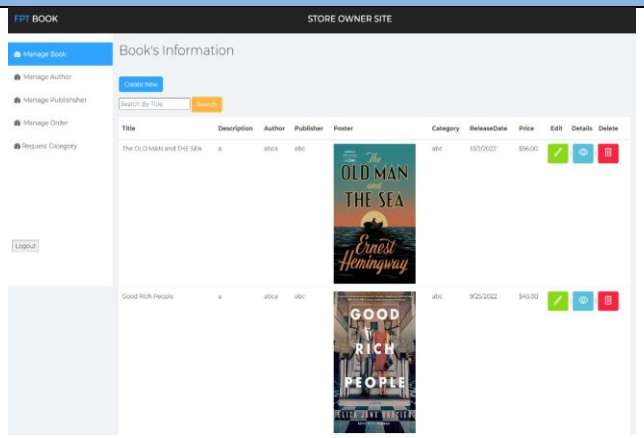
Customer Role				
1	Home page			The interface of the website has had some minor changes compared to the original wireframe design, these changes make the user interface more beautiful and smoother than before.

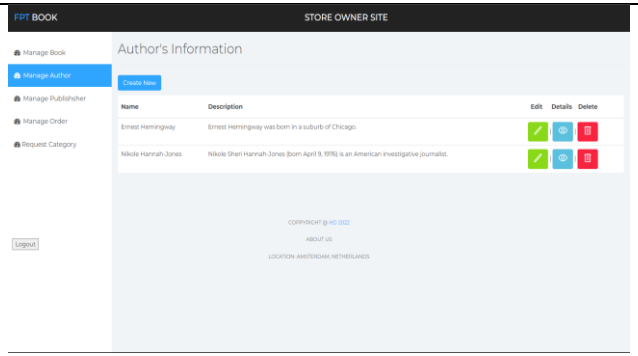
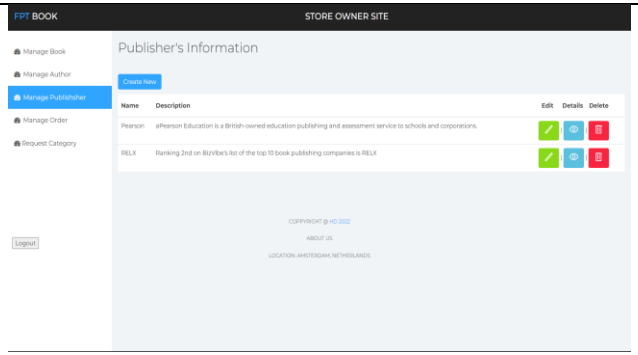
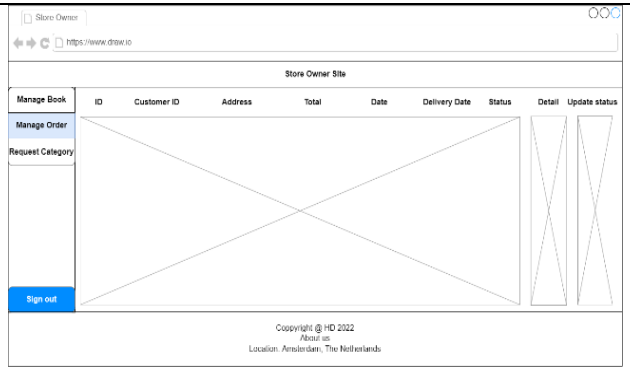
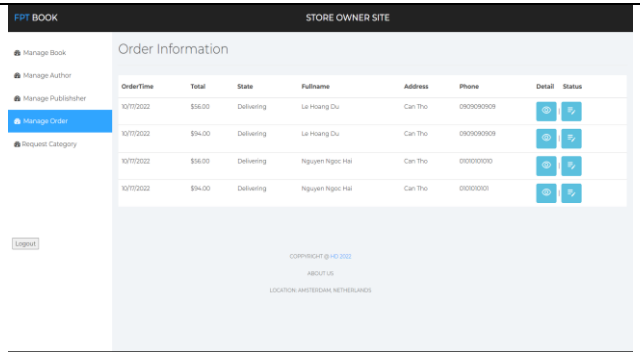
2	Help page			The interface of the website has had some minor changes compared to the original wireframe design, these changes make the user interface more beautiful and smoother than before.
3	About us page			Wireframe and website screenshots are mostly the same. The website interface is designed based on pre-design wireframes.

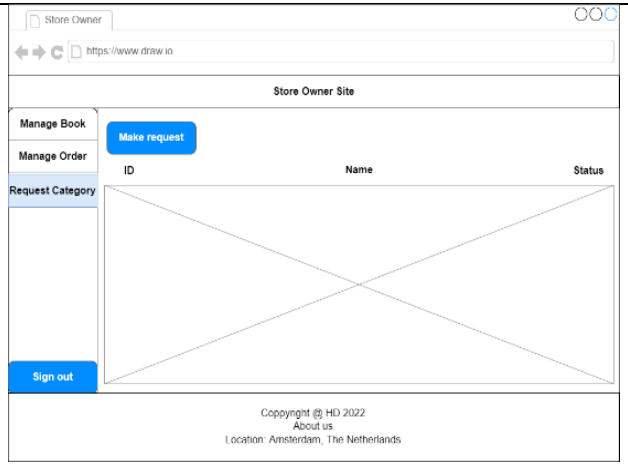
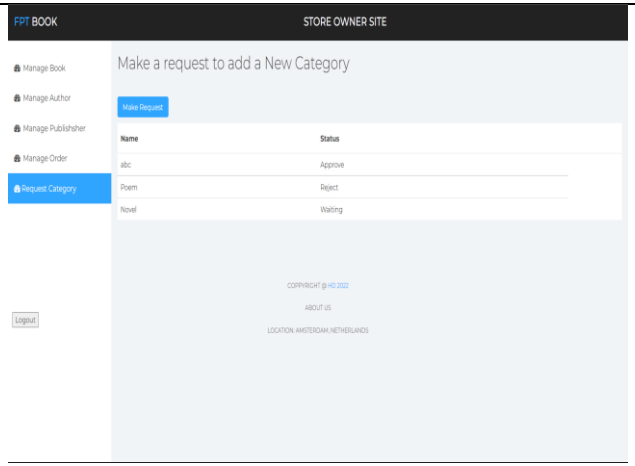
4	View Detail page			Wireframe and website screenshots are mostly the same. The website interface is designed based on pre-design wireframes.
5	Checkout page			Wireframe and website screenshots are mostly the same. The website interface is designed based on pre-design wireframes.

6	Profile page			The interface of the website after implementation was very different from the original wireframe design, this happened because during execution, the code had to follow a certain framework, so it was not possible to develop the web interface as before.
Admin Role				
7	Manage User Site			I merged customer management and shop owner management in one user management function to make it simple to handle. The bulk of website wireframes and

				screenshots are the same. The interface of the website was created using wireframes from the pre-design phase.
8	Manage Role Site			This site was added by me after reporting on my design (from Assignment 1)
9				Wireframes and most screenshots from websites are nearly identical. The website's interface was developed using wireframes from the pre-design phase.

10	Report Site			This site was added by me after reporting on my design (from Assignment 1)
Store Owner Role				
11				

12				This site and functionality was added by me after reporting on my design (from Task 1)
13				This site and functionality was added by me after reporting on my design (from Task 1)
14				Wireframes and most screenshots from websites are nearly identical. The website's interface was developed using wireframes from the pre-design phase.

15				Wireframes and most screenshots from websites are nearly identical. The website's interface was developed using wireframes from the pre-design phase.

Compare to the initial requirements

Table 3: Compare to initial requirements

No	Initial requirement	Done	Not done
Admin role			
1	Admin can sign in to the FPTBook system or sign out of the FPTBook system.	✓	
2	Admin can manage customer account such as viewing, adding, updating, or deleting customer account.	✓	
3	Admin can manage store owner account such as viewing, adding, updating, or deleting store owner account.	✓	
4	Admin can manage category such as approving or rejecting the new book category request made by the Store Owner.	✓	
Store Owner role			
1	Store Owner can sign in to the FPTBook system or sign out of the FPTBook system.	✓	
2	Store Owner can manage books such as adding, updating, searching, or deleting a book.	✓	
3	Store Owner can make a request to the Admin for adding the new book category.	✓	
4	Store Owner can manage orders such as viewing all records ordered by the customer, updating status, canceling an order.	✓	
Unauthenticated User role			
1	Unauthenticated user can sign up a new account.	✓	
2	Unauthenticated user can sign in to the FPTBook system.	✓	
3	Unauthenticated user can view the list of books.	✓	
4	Unauthenticated user can search a book.	✓	
5	Unauthenticated user can view detailed information about a specific book.	✓	
6	Unauthenticated user can view the list of help topics.	✓	
Customer role			
1	Customers can manage their account information by updating the Customer' full name, email, password, and deleting account.	✓	
2	Customers can view all their past orders.	✓	
3	Customers can view the list of books that they can buy.	✓	
4	Customer can view detailed information about a specific book.	✓	

5	Customer can perform the sequence of steps to buy a book such as searching book, viewing detail book, adding to cart, selecting quality, and paying.	✓	
6	Customer can view the list of help topics.	✓	

3.2 Conclude whether the application adapts all requirements or it needs to be improved later

Most of the functions stated in the SRS are performed (90%). Based on the project's peer review and requirement, several functions were added as well as removed, such as:

- Separate the interface to view the customer's orders to an independent interface from the profile interface.
- Change the way questions and answers are displayed in the Help interface.
- Change the display of categorical indexes in the homepage interface.
- The function of managing customers and store owners has been removed and replaced by user management to be able to manage data more easily.
- Added role management function, where admin can view, add and manage user roles.
- Add a report function, where the admin can see the statistics of the number of books, the total quantity and the total cost of goods sold, in addition, there is also an excel file export function so that the admin can easily view the statistics.
- Add an author management function, so that shop owners can manage the details of the authors of the books.
- Add a publisher management function, so that shop owners can manage the details of the publisher of the books.
- Use icons from Google Font, Font Awesome to improve user interface.

3.3 Analyse the factors that influence the performance of the application

- **Application Design**

Inefficient coding approaches at the application layer, subpar SQL queries, and poorly designed network hardware commonly slow down websites.

- **Third-Party Components**

Third-party web components have a significant impact on the performance of online applications. If the data from a third party becomes unavailable when the FPTBook system is reliant on it, it could not function. The system's performance may be affected by the poor performance of a third party.

- **Network**

The performance of an application can be severely harmed by increased latency, erratic contention, and irregular bandwidth. The network's erratic connection is one of the largest performance risks for most businesses; if the developer can't control how information is transmitted to the client, they can't control how

soon it arrives. Due to this, it is quite challenging to specify how the FPTBook online system developer should design it.

- **Database**

The database layer is at blame for around 90% of application performance issues (Data from Gleanster Research). This is due to a variety of factors, including the fact that databases constitute the foundation of all systems, their sophistication, and the difficulty of scaling them horizontally. When an application experiences a significant rise in demand, the database may in reality become a serious bottleneck. The database scales more slowly than the bulk of the other parts of the infrastructure. Scaling out—where the application utilizes several database servers concurrently to scale reads—is difficult at the application layer.

- **Mobile Environment**

Currently, more and more people are using APIs in general and web APIs in particular. Modern application architectures, which are becoming more dispersed and agnostic to programming languages, as well as the growing number of consumers engaging with software via mobile devices like smartphones and tablets, have driven the development of application APIs. Tablets and mobile apps (or mobile versions of desktop apps) must be developed with these platforms in mind for the best results.

- **The web browser**

The web browser is the key blind spot for acquiring true end-to-end visibility into the performance of online applications. Due to new approaches to application design and a growth in the usage of Web Services, it has become clear that one of the essential requirements for full visibility into application performance is the ability to trace the processing that takes place inside the browser.

- **Server/Hosting**

Your website's performance may be significantly impacted by the web host you choose. The approach you decide on is equally important. However, if you have a popular website that receives lots of traffic or if your website requires a lot of data for features like streaming videos or special applications, a shared host won't cut it. The RAM and bandwidth constraints will have a negative impact on the performance of your website. leading to a bad customer experience.

3.4 Evaluate the strengths and weaknesses of the application

3.4.1 The strengths of the application

- **UI Design**

It contains several built-in features thanks to the use of Bootstrap templates, which save you time by eliminating the need to write the necessary code. Some of the main benefits are responsive layouts, customizable form elements, typography, JS interaction, cross-browser compatibility, and more. The system's user interface is friendlier. Templates made using Bootstrap are adaptable and simple to change. You may either keep it as is or add some of your own features based on your needs. Users of this website can easily get the necessary data.

- **User-focused on interface**

The content and buttons are organized in the right order to avoid overly diverting the user. Display of accessible information on users, books, etc. is clear and thorough. The placement of the function buttons is straightforward and thorough, and it adheres to the original design.

- **Code clean**

All source code and functions are clearly distinguished so that developers can easily detect and correct errors

```
// POST: Categories/Edit/5
// To protect from overposting attacks, enable the specific properties you want to bind to.
// For more details, see http://go.microsoft.com/fwlink/?LinkId=317598.
[HttpPost]
[ValidateAntiForgeryToken]
0 references
public async Task<IActionResult> Edit(int id, [Bind("Id,Name,Status")] Category category)
{
    if (id != category.Id)
    {
        return NotFound();
    }

    if (ModelState.IsValid)
    {
        try
        {
            _context.Update(category);
            await _context.SaveChangesAsync();
        }
        catch (DbUpdateConcurrencyException)
        {
            if (!CategoryExists(category.Id))
            {
                return NotFound();
            }
            else
            {
                throw;
            }
        }
        return RedirectToAction(nameof(Index));
    }
    return View(category);
}
```

```
// GET: Categories/Delete/5
[Authorize(Roles = "Admin")]
0 references
public async Task<IActionResult> Delete(int? id)
{
    if (id == null || _context.Category == null)
    {
        return NotFound();
    }

    var category = await _context.Category
        .FirstOrDefaultAsync(m => m.Id == id);
    if (category == null)
    {
        return NotFound();
    }

    return View(category);
}

// POST: Categories/Delete/5
[HttpPost, ActionName("Delete")]
[ValidateAntiForgeryToken]
0 references
public async Task<IActionResult> DeleteConfirmed(int id)
{
    if (_context.Category == null)
    {
        return Problem("Entity set 'FPTBookContext.Category'  is null.");
    }
    var category = await _context.Category.FindAsync(id);
    if (category != null)
    {
        _context.Category.Remove(category);
```

Figure 73: Code clean

- **Images and Font**

Font Awesome is the font that is used on the system since all the icons are included in a single font file and loading Font Awesome just requires one HTTP request. This has great performance benefits, therefore you should consider utilizing Font Awesome as a remedy. Using Font Awesome also has the benefit of being easy to read and see.

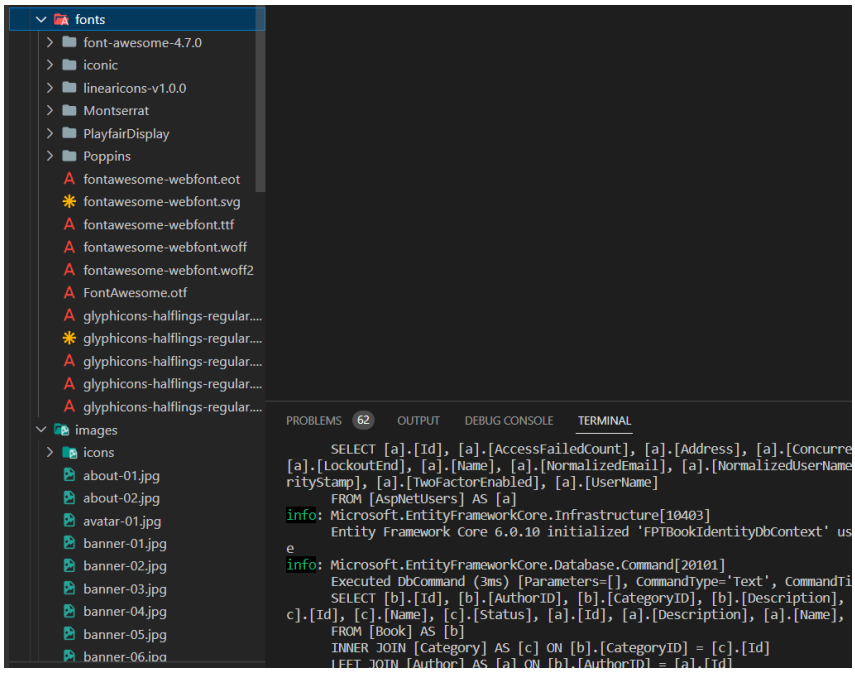


Figure 74: Images and Font used in our website

- **Decentralization**

The developer considered a wide range of scenarios and potential outcomes that may occur during user use and affect the functionality of the software. Regarding decentralization, the developer has worked with a range of situations to protect information as expertly as is practical while avoiding unexpected blunders.

- **Stability and interlinking database**

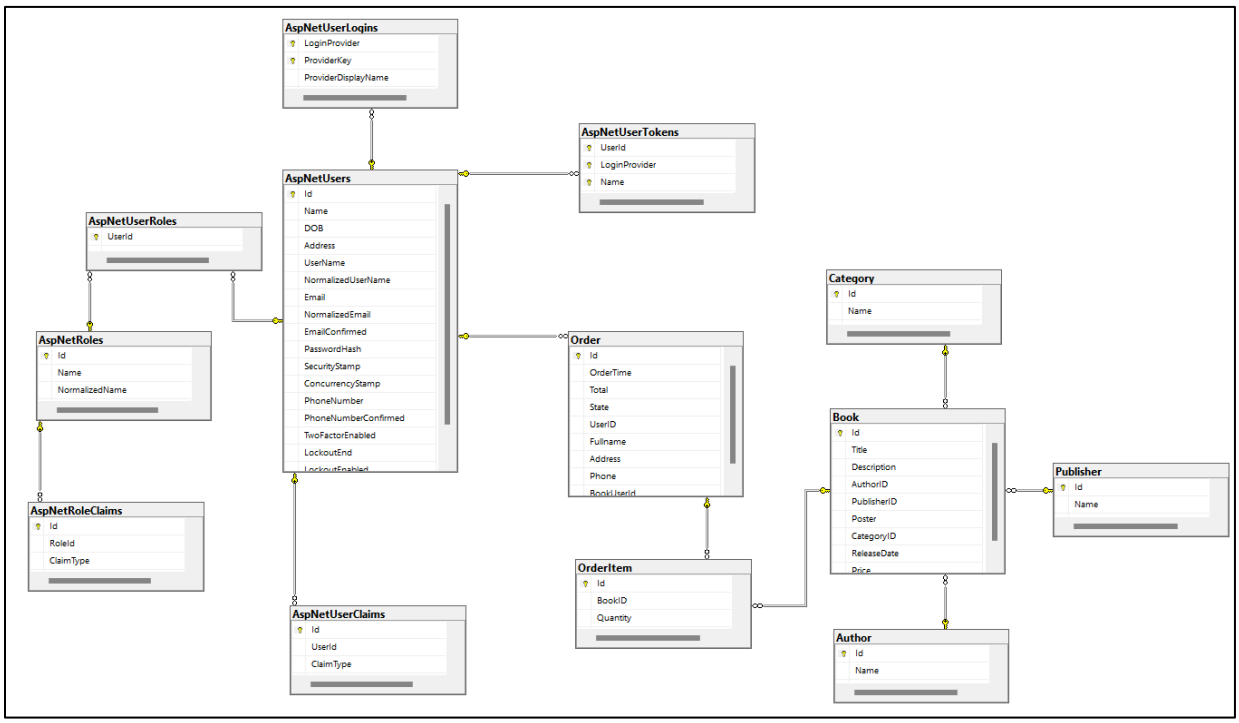


Figure 75: Database of application

- **User interface and Functions**

Intuitive user interface, easy to use, functions classified by different roles

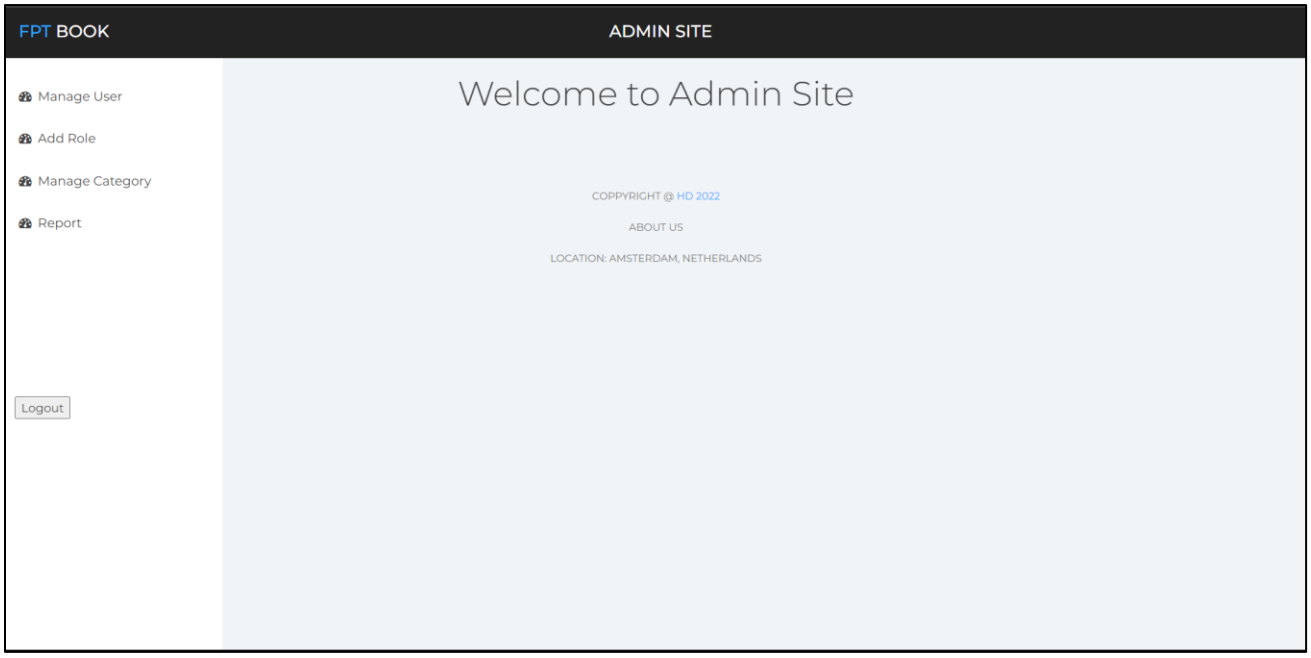


Figure 76: Admin role interface

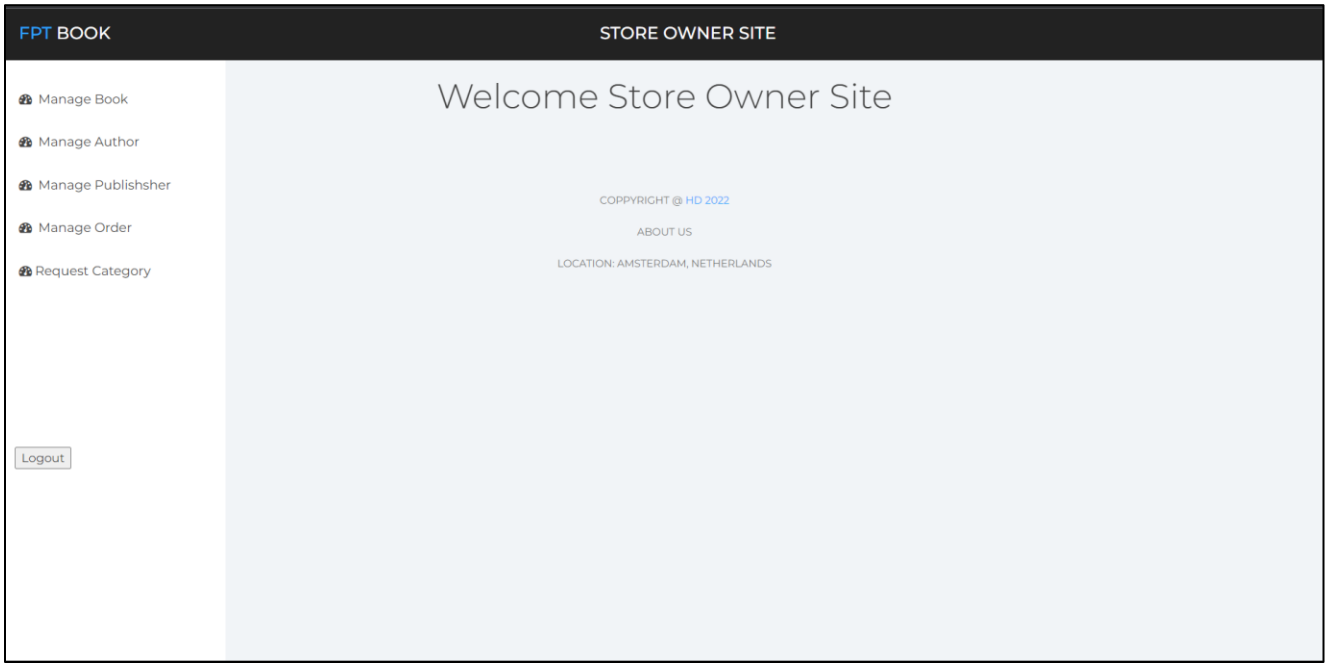


Figure 77: Store owner role interface

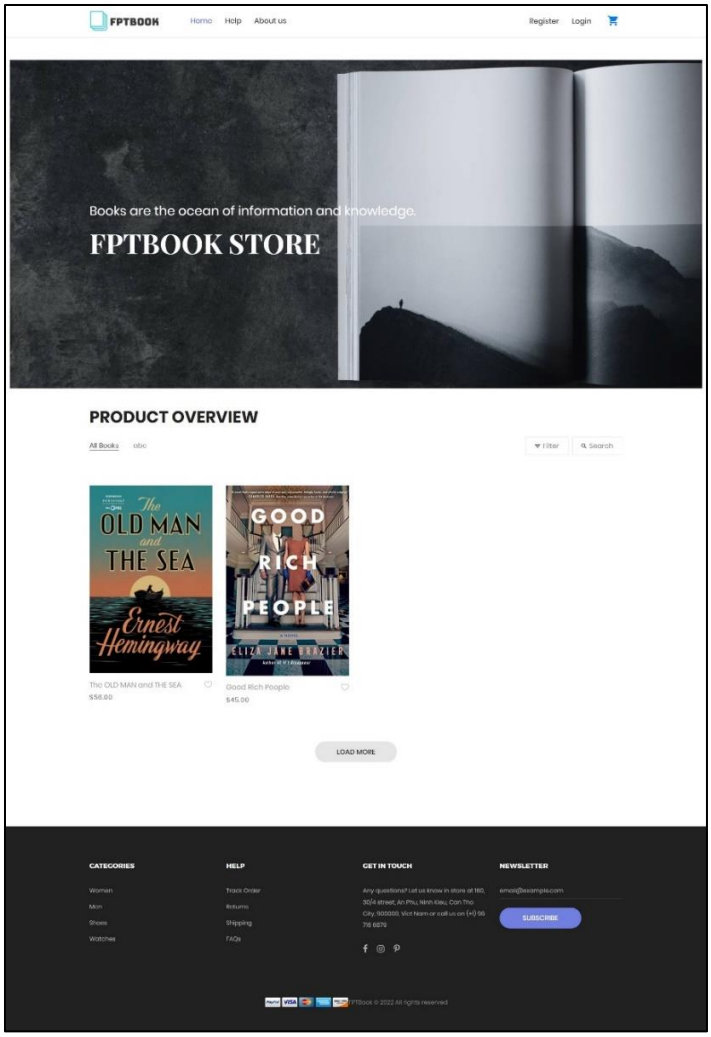


Figure 78: Customer role interface

The admin role has a revenue report function and can export to an excel file

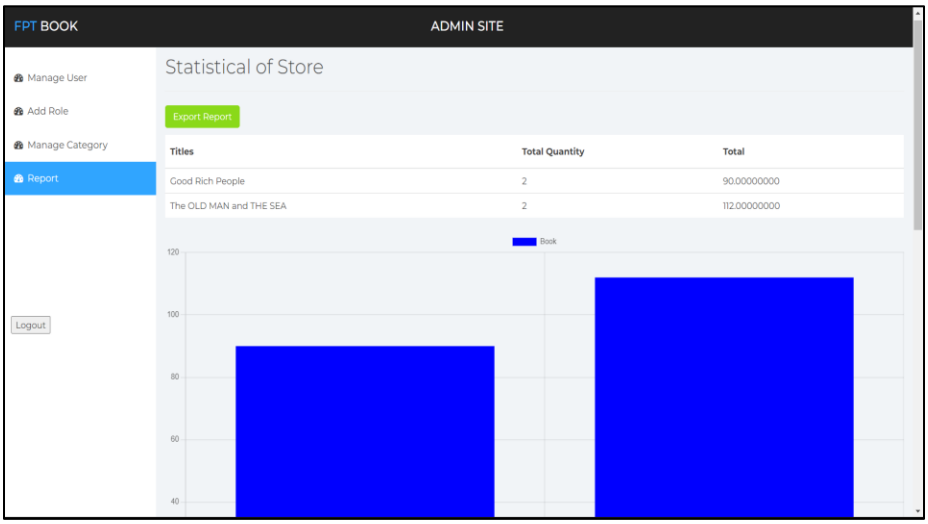


Figure 79: Report function of Admin role

Internal operations and the primary information, including reading data, generating, changing, and deleting information, are made to be simple to see and utilize. Customers don't need to learn anything; they can just browse an intuitive, easy-to-understand website with a clear website design and page structure.

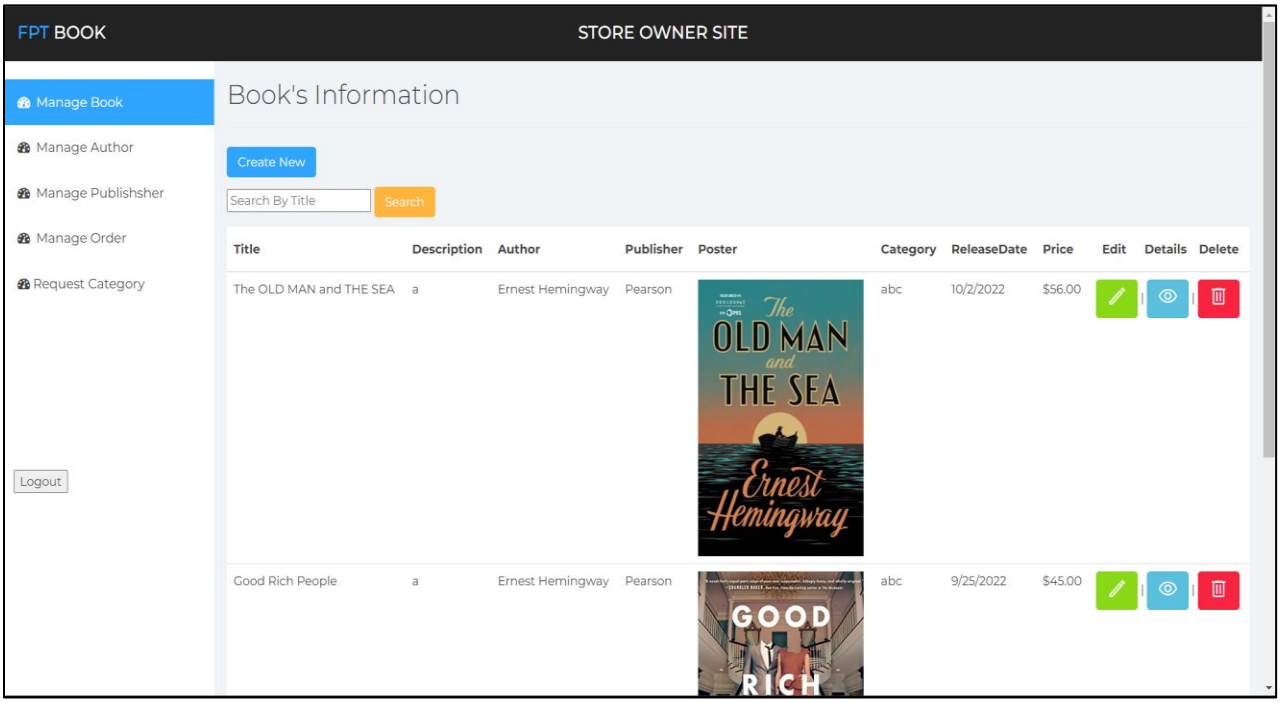


Figure 80: Viewing data, creating, updating, updating and deleting information functions

The option to choose the contents of the Roles is shown on the left side of the screen, users are easy to use and manage what features they are using.

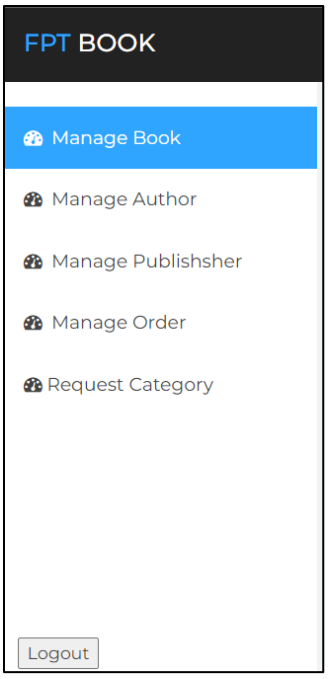


Figure 81: Options to choose the functions

Mobile optimized pages

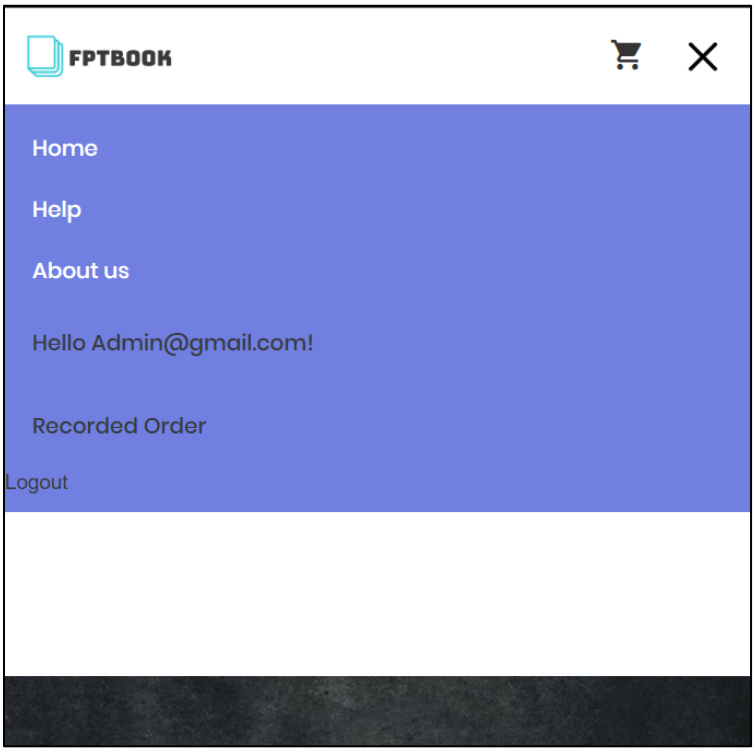


Figure 82: Mobile optimized navigation bar

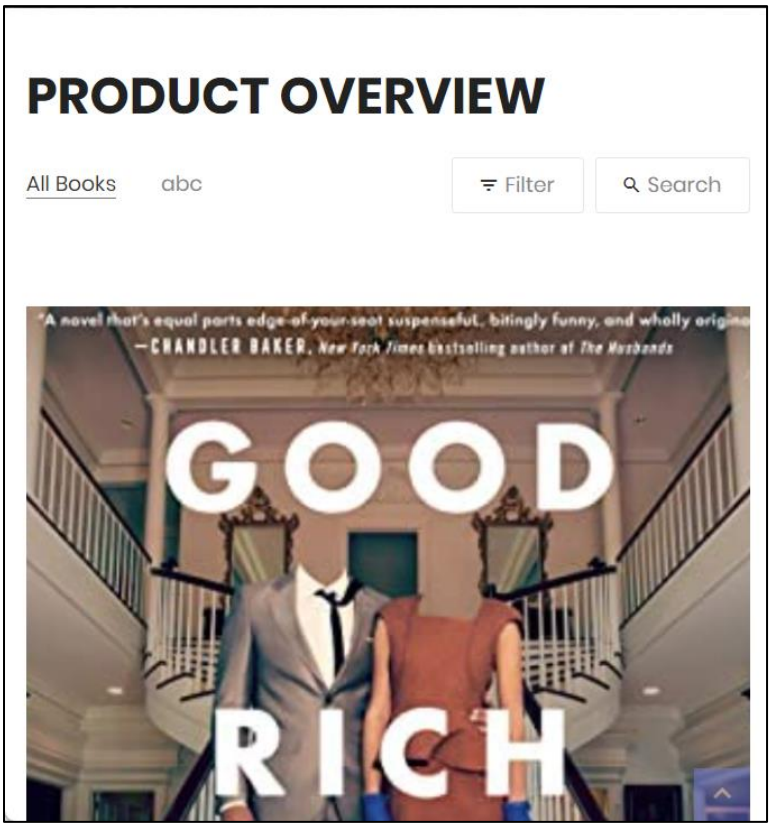


Figure 83: Mobile optimized home page

- **Security**

The user's password is encrypted before being put into the database

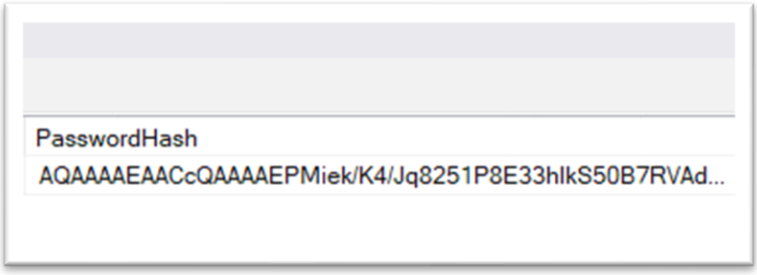


Figure 84: Password is encrypted in the database

Functional hierarchy exists. Only the features that the system permits can be used by each user. The system prohibits any attempts to get access, and it will alert users who are not permitted to utilize a function.

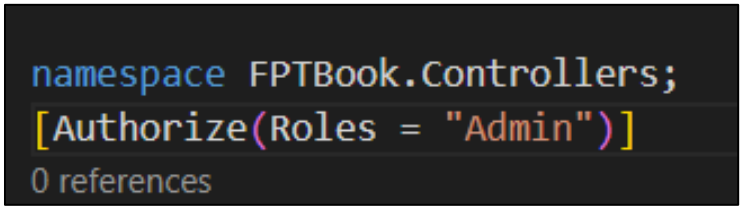


Figure 85: Authorize Admin role

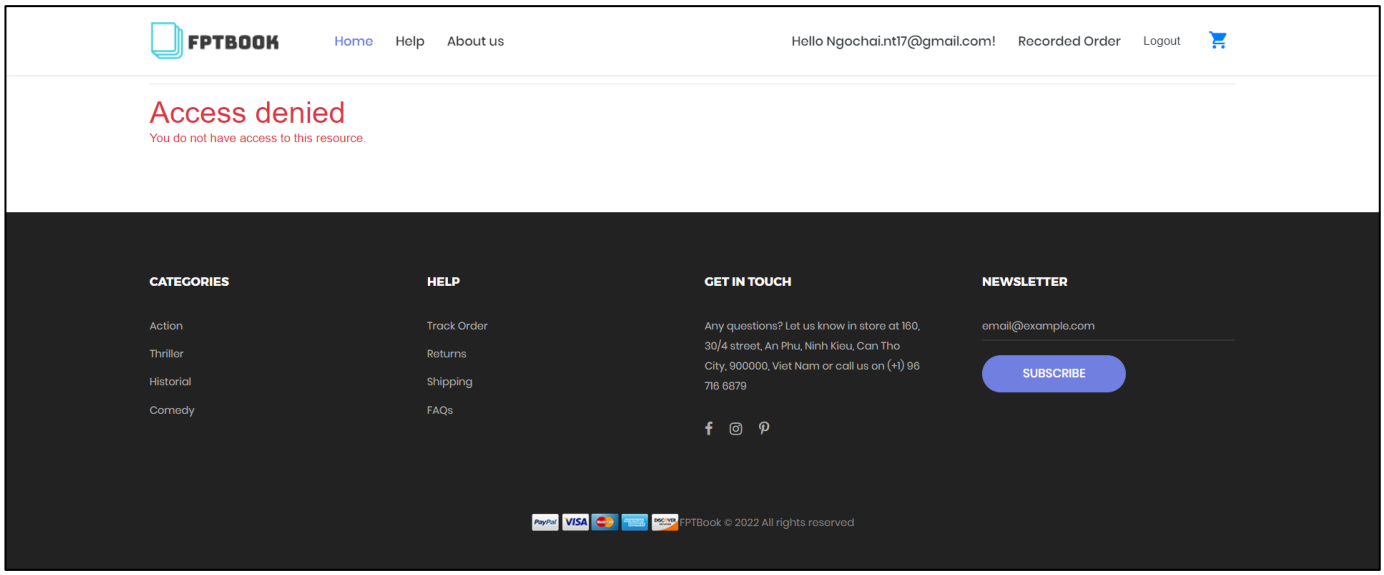


Figure 86: The system will notify if trying to access with roles other than administrator

3.4.2 The weaknesses of the application

- **UX Design**

Because it employs the default version of Font Awesome, the system is limited in the amount of icons it can use. Since certain icons must employ corresponding icons as a consequence, the system's buttons and actions are not represented by them.

- **Interactive**

There is no notification function for users, especially customers to keep information about orders. Moreover, the user's profile picture is not available. Additonally, no chat box for communication between shop owner and customer.

- **Implement code**

Database data retrieval requires much too many queries.

```
0 references
public IActionResult Create()
{
    return View();
}

// POST: Authors/Create
// To protect from overposting attacks, enable the specific properties you want to bind to.
// For more details, see http://go.microsoft.com/fwlink/?LinkId=317598.
[HttpPost]
[ValidateAntiForgeryToken]
0 references
public async Task<IActionResult> Create([Bind("Id,Name,Description")] Author author)
{
    if (ModelState.IsValid)
    {
        _context.Add(author);
        await _context.SaveChangesAsync();
        return RedirectToAction(nameof(Index));
    }
    return View(author);
}
```

Figure 87: Create author function

- **Share on social media**

The website is not linked to other popular social networks such as Facebook, twitter, etc.

3.4.3 The improvement of the application

- **Boost speed:**

Site speed significantly affects conversion rates and bounce rates because consumers are impatient. Every second, more of your visitors will bounce, meaning they will close a tab or click the back button on their browser to leave your site before it has even had a chance to load. load. This applies particularly to online stores. For the greatest first-time visitor experience, increase the website's speed. Things you can do to increase website speed are:

  - **Clean up your code**

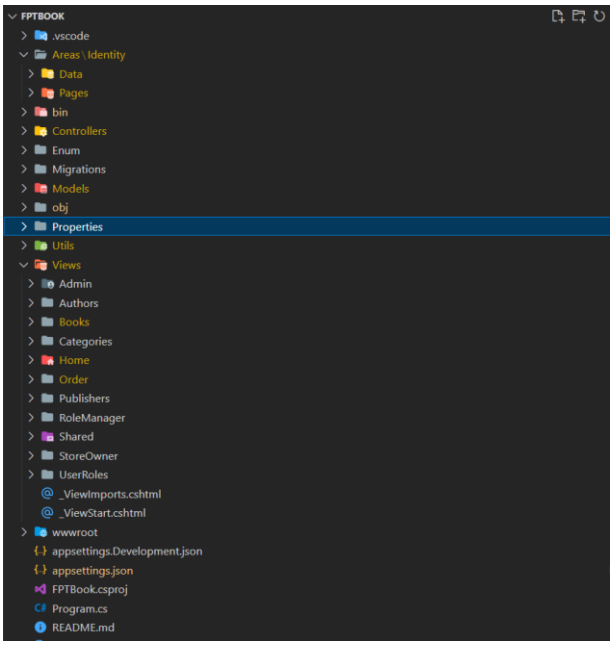


Figure 88: Folder Structure

- **Reduce image size:** The website interface would not be complete without images. But this is also what makes your website "heavy." As a result, you must balance picture size and quality, "optimizing" or retouching the image to obtain a high-quality image at the smallest size. The speed of the website will increase as a consequence.
- **Database optimization:** Through assisting plugin applications, the database may be optimized. A database that has been optimized will hasten the loading of websites. Installing and using the WP-Optimize plugin is a smart idea.
- **Optimize JS and CSS:** The WP Minify Fix plugin is a recommendation not to be overlooked for JS and CSS optimization. As a result, when you enable this plugin on your website, JS and CSS files will be combined and compressed to speed up page loading. This results in a quicker loading time.

- **Improve UX design:** Users may use Smooth Login to sign in with other related accounts, such as Google, Twitter, etc., and to use autofill to have their saved information automatically filled in throughout the login process. This makes the system more seamless.
- **Implement notification:** To user could be get message from system

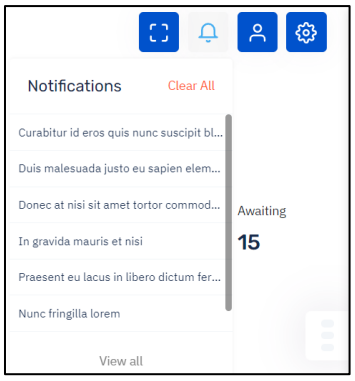


Figure 89: Notification interface

- **Implement user's profile picture:** For a more intuitive user interface

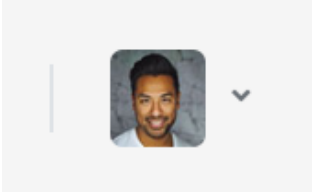


Figure 90: User's avatar

- **Implement chat box:** Here customers can chat with shop owners about their questions and needs

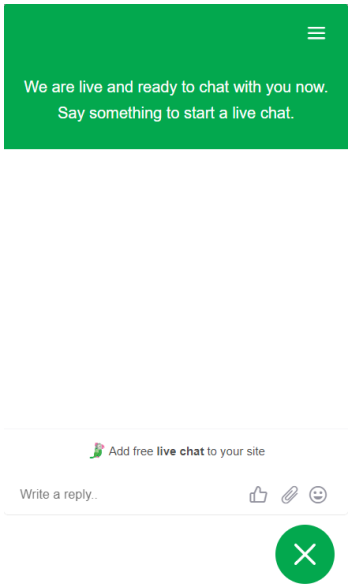


Figure 91: Chat box interface

- **Implement link with social medias**

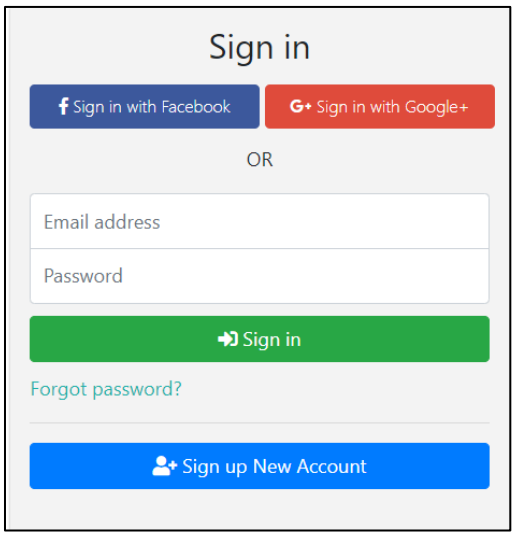


Figure 92: Link with social medias example

4 References

University Of Greenwich Learning Materials, 2022. *University Of Greenwich.* [Online]
Available at: https://flm.greenwich.edu.vn/gui/Home
[Accessed 21 October 2022].